

Subjectivity-Intersection Braid Quantum Gravity

A Constructive Refutation of Objectivist Primacy from Source-Derived Quantum Dynamics to the Einstein Limit

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Claim status. This is a mathematically explicit *quantum-gravity candidate*, not a completed or empirically confirmed theory. In one fixed pointed-Braid model, the source supplies an operational four-direction event cylinder, a Lorentz carrier, a scoped local recorded CTP parent, a four-component history-dressed total Ward identity, and one coefficient-free finite matter-to-geometry-to-matter feedback cycle. Independently, the declared long-wavelength, local, second-order, formally self-adjoint, conservative class yields $G_{\mu\nu} = (3/5)T_{\mu\nu}$ and a massless spin-2 sector with two physical helicities. The finite Fisher parent outputs relative coefficient 1, whereas 3/5 is the distinct full-corner routing ratio; their physical finite-to-continuum identification is not derived. Repeated nonlinear closure, gravitational first-class constraints, calibrated Newton coupling, realistic Standard Model matter, ultraviolet completion, and empirical validation remain open.

Abstract

We develop Subjectivity-Intersection Braid Quantum Gravity as a candidate theory in which a public spacetime carrier is generated by intersections of nonidentical, standpoint-bearing Braid processes rather than assumed as an ontologically prior container. Subjectivity here denotes a located mode of access, transport, or presentation, not consciousness. The construction begins with one pointed quantum-algebraic Braid source and maintains an explicit separation between finite source theorem, continuum limit, physical interpretation, and empirical claim.

The fixed source produces a permutation-covariant rank-four event moment

$$Q_4 = \frac{12}{25}P_0 + \frac{52}{25}P_1$$

and a source-null causal reflection $J_{\text{ray}} = -P_0 + P_1$. Their product $K_{\text{evt}} = J_{\text{ray}}Q_4$ has exact signature $(3+, 1-)$. The same marked event structure selects the full matter corner, ten independent response directions, fixed-model matter routing, and the relative normalization 3/5 without manual insertion. A refinement clock fixes the Euclidean event-semigroup rate $r = 2\pi/3$ and selects the Lorentzian infrared principal tensor by source continuation rather than a Wick-sign declaration.

The v0.8 advance was a controlled passage from this carrier to a continuous tensor and then to sourced dynamics. Eight typed source sectors carry exact rational $GL(4)$ transport and finite nonmetricity. Braid-derived Gibbs tangents extend them to a smooth event cylinder; the metric, its tangent, and nonmetricity glue over all 24 source charts and transform covariantly under passive local $GL(4)$. Event-path Fisher/Bogoliubov–Kubo–Mori horizontality uniquely lifts metric change to finite positive Markov rates. The lift is flat on all 45 metric two-planes and integrates locally to an exponential leaf. A source-product typing condition then removes the remaining solder ambiguity. Finally, the co-moving causal reflection

$$J(Q, \tau) = I - 2 \frac{(Q\tau)\tau^T}{\tau^T Q \tau}, \quad K(Q, \tau) = Q - 2 \frac{(Q\tau)(Q\tau)^T}{\tau^T Q \tau}$$

restores all ten Lorentz-metric variations without introducing an independent clock coupling. In the declared long-wavelength, local, second-order, formally self-adjoint, conservative class, the finite Braid action variation converges to Hilbert variation, yields an off-shell Noether identity and on-shell Ward conservation, and combines with the source-native first-order gravity sector to give

$$G_{\mu\nu}(g) = \frac{3}{5}T_{\mu\nu}(g, \lambda).$$

Its flat quadratic limit is the massless Fierz–Pauli gauge class with two physical helicities in four dimensions.

Version 0.81 adds a finite source-cylinder completion in a narrower, explicitly typed class. The source-fixed raw/Petz record contrast has conditional effect tangents $+D_\alpha/2$ and $-D_\alpha/2$, so its gauge-invariant Choi marginal difference is the complete four-transfer defect $D_\alpha = S_\alpha - \Phi(S_\alpha)$. Placed cellwise on the operational cylinder, this gives a block-local 3 + 1 recorded CTP parent with CPTP normalization, retarded support, positive noise, and exact address/time refinement. The same D_α is the interaction-stress compression in the strong collision intertwiner and yields the exact causal Ward telescope

$$\sum_{k=0}^{n-1} \Phi^k(D_\alpha) = S_\alpha - \Phi^n(S_\alpha)$$

for all four components and all eight source sectors. A full-rank record Fisher action then supplies one finite identity-feedback cycle from matter record to metric register and back to a CPTP matter channel without MMR, CGR, an Einstein target, a manually inserted 3/5, or fitted coefficients.

The result remains scoped. The finite parent uses the aligned four-score and block-local collision class rather than full ten-component nonlinear metric dynamics. Identity feedback is the minimal type-preserving closure, not a uniqueness theorem over all dynamics, and repeated nonlinear rank preservation is unproved. The direct Fisher parent has relative coefficient 1; the earlier 3/5 remains a source-derived full-corner routing ratio but is not the same coefficient. Source-only absolute units are nonidentifiable, the current internal matter algebra is not Standard-Model equivalent, and the finite internal Lie algebra does not yet supply gravitational first-class constraints. The results refute the universal necessity thesis of *objectivist primacy* by construction; they do not refute mind-independent reality, establish phenomenal consciousness, or show that nature realizes the model.

Keywords: Subjectivity-Intersection Braid Quantum Gravity; quantum gravity candidate; pointed braid; generated objectivity; objectivist primacy; Lorentz signature; emergent spacetime; Einstein equation; metric-affine gravity; Ward identity.

1 What changed from v0.8

Version 0.8 derived the Einstein equation in a declared infrared universality class but left the finite Lorentzian parent, interaction stress, four-component discrete Ward balance, and finite backreaction open. Version 0.81 resolves the first three and constructs one finite backreaction cycle inside the aligned four-score, block-local source-cylinder identity-feedback class. It also separates two previously conflated coefficients: the direct finite Fisher coupling is 1, while 3/5 is the independently derived full-corner routing ratio used by the scoped continuum equation.

The strongest defensible claim is upgraded but narrowed with greater precision. The former finite-parent and four-current gaps are closed in one declared source-cylinder class. The unrestricted SDPC claim is not declared solved because full ten-component nonlinear dynamics, a source-forced unique feedback law, repeated evolution, gravitational constraint closure, all-scale control, physical calibration, and empirical identification remain incomplete.

2 Subjectivity intersection and the target of the refutation

2.1 Subjectivity is not consciousness

A *subjectivity* in this work is a typed and located mode of access, transport, or presentation whose distinction from other such modes is constitutive of the construction. It may be represented by a source prefix, a reference system, or an oriented transport context. No braid, photon, clock, algebra, or source sector is thereby attributed sensation, awareness, intention, or phenomenal experience. Any bridge from this formal notion to consciousness would require a separate axiom and separate evidence.

Question	v0.81 result	Remaining boundary
Operational four-direction carrier	Retained exactly from the source cylinder and four null links	No universal four-dimensional uniqueness theorem
Matter realization and 3/5	Fixed-model MMR2 closed; normalized source-cup ratio derived	Arbitrary instruments and SI-unit calibration remain open
Lorentz quantum parent	Local raw/Petz-recorded CTP parent on the source cylinder	Aligned four-score and block-local scope; full ten-component nonlinear parent open
Continuous tensor field	S4 gluing and passive local $GL(4)$ naturality exact	Empirical identification with natural spacetime remains open
Physical stress and Ward law	Branch contrast derives all four transfer defects; exact history-dressed finite Ward	Arbitrary smooth diffeomorphism Ward and spatial-gradient dynamics open
Einstein and spin-2 limit	Noncircular scoped derivation with two helicities	Direct finite coefficient 1 is not yet identified with routing ratio 3/5
Finite backreaction	One parameter-free CPTP matter–geometry–matter cycle	Identity-feedback uniqueness and repeated nonlinear closure open
Physical completion	Two external calibration anchors are sufficient in class	Actual anchors, Newton constant, realistic matter, and measurements absent

Table 1: The v0.81 dependency split. “Derived” always means derived inside the declared fixed source model and stated class, not established as a universal law of nature.

Let A and B denote nonidentical source standpoints. Their intersection does not mean a third external observer that views both from nowhere. It is the compositional event in which their difference is retained while presentation redundancy is removed. The public structure $C(A, B)$ is then an output of the intersection rather than an independently completed container.

Definition 2.1 (Generated objectivity). A carrier $C(A, B)$ exhibits generated objectivity when it is invariant under allowed internal presentation changes, covariant under ambient re-description, symmetric under exchange of the two typed standpoint roles when the source law requires it, altered by the deletion of either role, and regenerated by one source law through the declared refinement stages without stage-dependent fitting.

2.2 The exact objectivist claim being denied

Definition 2.2 (Objectivist Primacy Thesis, OP). Every mathematically admissible construction of a stable, shareable, transformation-respecting physical carrier must take that carrier, or an equivalent perspective-neutral carrier, as primitive. Distinct standpoints may describe it, but cannot generate it.

OP is a universal necessity thesis. It is stronger than scientific reproducibility, realism about measured regularities, covariance, or mind-independent existence. A single valid generated-objectivity model refutes OP as stated. It does not prove that the model is realized in nature. The present paper therefore keeps two conclusions separate:

1. a mathematical counterexample to the necessity of objectivist primacy; and
2. a physical quantum-gravity hypothesis whose empirical truth remains open.

The proposal is adjacent to, but not identical with, relational quantum mechanics, quantum-reference-frame programmes, QBism, perspectival realism, and structural realism [11, 5, 14, 3, 10, 8, 15]. Its distinct technical claim is generative: a shared tensor carrier is reconstructed from maintained source provenance and intersection rather than presupposed as a perspective-neutral starting object.

3 Pointed-Braid source and operational event cylinder

3.1 Quantum-algebraic source

At refinement stage n , take the matrix algebra

$$\mathcal{A}_n = M_{5^n}(\mathbb{C}), \quad j_n(X) = X \otimes I_5. \quad (1)$$

The source contains distinguished states, marked local algebras, and unitary Braid transports satisfying the typed Yang–Baxter and composition laws. This is quantum in the algebraic sense: states, complex noncommutative observables, tensor composition, and unitary transport are present. The present paper does not derive complex amplitudes, the Born rule, or Hilbert-space composition from an unstructured topological braid group.

The source data are therefore neither “bare braid topology” [1] nor a continuum spacetime. They are a finite quantum-algebraic pointed-Braid source developed in the preceding Subjectivity-Intersection Mathematics programme [16, 17]. The scientific question is how much geometry and dynamics this source forces without target fitting.

3.2 Operational event identification

The fixed source packet contains a marked event effect P and a spectral event algebra. The actual stored event transition R_{evt} acts on 25 local atoms, is involutive, and satisfies the declared Yang–Baxter event relation. The same marked generator determines four future-null affine links. This identification retires an independent formal correspondence principle CGR inside the fixed model: the event carrier is read from the source law itself rather than attached as a separate map.

This is a scoped statement. It does not say that every operational event in nature must be represented by this P , nor does it select a natural signed sector. It says that the previously separate formal event correspondence is redundant in the declared source construction.

3.3 Four dimensions without a primitive completed spacetime

The operational cylinder has four source directions because four independently certified null links span the event carrier. Let L be the matrix with these links as columns. The exact certificate gives

$$\det L = \frac{81}{16} \neq 0. \quad (2)$$

Thus the event carrier is four-dimensional in the fixed model. This is stronger than postulating an arbitrary ambient $\mathbb{R}^4$ and then placing Braid data inside it: the four operational directions are generated and are nondegenerate. It is weaker than a theorem that four dimensions are inevitable for all admissible Braid sources. The latter is not claimed.

4 Matter routing and the non-manual factor 3/5

4.1 The full event-conditioned corner

The marked event defines the physical corner candidate

$$\mathcal{A}_\cap = PAP, \quad (3)$$

with Hamiltonian $H_\cap = PG_1P$ and ten compressed source operators

$$F_A^\cap = PF_AP, \quad A = 1, \dots, 10. \quad (4)$$

The corresponding Gibbs family is faithful on the corner, and its Bogoliubov–Kubo–Mori Hessian has rank ten. Hence the ten declared matter-response directions are locally nondegenerate.

The source atoms generate five partition retractions. Requiring exact preservation of the already fixed eleven-operator packet

$$\mathcal{S}_\cap = \{PG_1P, PF_1P, \dots, PF_{10}P\} \quad (5)$$

selects only the full partition. Every nontrivial dephasing changes $H_\cap$ before any coefficient can be retuned. This closes MMR2 within the declared fixed event-conditioned model and complete five-partition class. It does not prove uniqueness among all completely positive instruments or all possible subalgebras.

4.2 Where 3/5 comes from

The normalized tail trace sends the corner unit to the outer event effect P . Its source-cup composition yields the relative multiplier

$$\kappa_B = \frac{3}{5}. \quad (6)$$

No line of the Lorentz-carrier or backreaction calculation sets or retunes this value. It is inherited from the same marked source routing that selects the event-conditioned matter packet. In particular, the appearance of the same coefficient in the Einstein equation is not counted as proof of that equation: source-derived normalization, tensor typing, conservation, and common metric variation are proved as separate gates.

5 Braid-derived local-label dynamics

5.1 Actual event transition versus identity and flip

For each adjacent pair of source directions, define the displacement under the actual Braid event transition. Its exact pair moment is

$$M_{\text{pair}} = \begin{pmatrix} 28 & -20 \\ -20 & 28 \end{pmatrix}, \quad (7)$$

with eigenvalues 8 and 48 and rank two. The identity comparator has rank zero and the ordinary flip comparator rank one. Thus the rank-two coupling is specific to the actual non-flip event map within this comparator class.

Average the adjacent-pair moment over the 24 source charts. The result is

$$Q_4 = \frac{104I_4 - 20\mathbf{1}\mathbf{1}^\top}{50} = \frac{12}{25}R_0 + \frac{52}{25}R_1, \quad (8)$$

where

$$R_0 = \frac{1}{4}\mathbf{1}\mathbf{1}^\top, \quad R_1 = I_4 - R_0. \quad (9)$$

Q_4 has diagonal entries $42/25$, off-diagonal entries $-2/5$, and eigenvalues $12/25$ on the uniform line and $52/25$ on its three-dimensional complement. It is positive and full rank. No distinguished physical $1 + 3$ ordering or external axis weights enter Eq. (8).

5.2 Relative entropy and the BKM–Dirichlet form

The source response is the derivative of a log-partition function. Within the response-exact Bregman class this selects the log-Z Bregman divergence, equivalently the Umegaki relative entropy for the Gibbs family [13, 2]. On actual neighbor event edges, the quadratic term is

$$\delta^2 S_{\text{edge}} = \frac{1}{2} \int Q_4^{ab}, \langle \partial_a \lambda, \partial_b \lambda \rangle_{\text{BKM}}, d\mu. \quad (10)$$

The base moment has rank four and the internal BKM metric rank four in the tested local-label sector, so the gradient quadratic form has rank sixteen. This supplies source-independent local-label dynamics on the fixed cylinder. It is a positive Dirichlet form, not yet a Lorentzian action.

6 Null tetrad and source-derived Lorentz carrier

6.1 The source-null metric

The four null links have a source-fixed Gram matrix G_{null} with zero diagonal and off-diagonal entries $-12/25$. Since L is invertible, the ambient metric reconstructed from the links is unique:

$$g = L^{-\top} G_{\text{null}} L^{-1} = \text{diag}\left(-\frac{16}{25}, \frac{16}{75}, \frac{16}{75}, \frac{16}{75}\right). \quad (11)$$

It has signature $(3+, 1-)$. The induced map on symmetric tensors, $\text{Sym}^2(L)$, has rank ten. Thus the four-direction carrier is algebraically capable of supporting all ten symmetric rank-two components.

This statement alone does not solve the physical stress problem. Full rank is an existence fact about an algebraic map; it does not prove that the actual matter operator packet transforms through that map or that the resulting response is conserved.

6.2 Why a coframe alone cannot create the Lorentz sign

The positive event moment remains positive under the null coframe:

$$LQ_4L^\top = \text{diag}\left(\frac{27}{25}, \frac{117}{25}, \frac{117}{25}, \frac{117}{25}\right). \quad (12)$$

By Sylvester’s law of inertia, no real invertible congruence changes signature $(4+, 0-)$ to $(3+, 1-)$. Therefore a claim that “the null tetrad automatically Lorentzianizes the positive edge action” is false. This no-go forces the causal sign to have a source origin other than coordinate change.

6.3 Causal grading

The source causal analysis distinguishes the uniform sum of the four future-null rays as the timelike direction. The corresponding reflection is

$$J_{\text{ray}} = I_4 - \frac{1}{2} \mathbf{1}\mathbf{1}^\top = -P_0 + P_1. \quad (13)$$

It has eigenvalue -1 on the uniform line and $+1$ on the standard three-dimensional complement. Since Q_4 is S_4 -equivariant, J_{ray} and Q_4 commute. Hence

$$K_{\text{evt}} = J_{\text{ray}} Q_4 = -\frac{12}{25} P_0 + \frac{52}{25} P_1 \quad (14)$$

is symmetric with exact signature $(3+, 1-)$. In components its diagonal is $36/25$ and every off-diagonal entry is $-16/25$.

Theorem 6.1 (Source-derived Lorentz event carrier). *In the declared fixed source model, the actual Braid event moment Q_4 and the independently source-selected causal grading J_{ray} determine the Lorentz-signature carrier $K_{\text{evt}} = J_{\text{ray}} Q_4$. The construction uses no Wick rotation, manual sign assignment, or fitted continuous coefficient.*

6.4 Natural normalization to the null Gram tensor

Because Eq. (14) is contravariant in the kinetic form, its physical comparison target is G_{null}^{-1} , not G_{null} . Restrict a normalizer to the positive, self-adjoint, S_4 -equivariant class

$$A_{\text{kin}} = a_0 P_0 + a_1 P_1. \quad (15)$$

Then

$$A_{\text{kin}}^T K_{\text{evt}} A_{\text{kin}} = G_{\text{null}}^{-1} \quad (16)$$

fixes

$$a_0 = \frac{25\sqrt{3}}{36}, \quad a_1 = \frac{25\sqrt{39}}{156} \quad (17)$$

uniquely within that class. All 24 chart permutations commute with the projector decomposition. In the character frame one recovers

$$g^{-1} = \text{diag}\left(-\frac{25}{16}, \frac{75}{16}, \frac{75}{16}, \frac{75}{16}\right). \quad (18)$$

The uniqueness claim is deliberately local: it is not uniqueness under arbitrary Lorentz isometries or among all nonlinear field redefinitions.

7 From a Lorentz carrier to a Lorentz action

7.1 Why the action question is separate

The positive Umegaki edge functional underlying Eq. (10) has Hessian $Q_4 \otimes G_{\text{BKM}}$. The causal carrier has principal tensor $K_{\text{evt}} \otimes G_{\text{BKM}}$. Since the latter contains one negative mode, equality of the two cannot be obtained by a real invertible coframe congruence. A carrier identity is therefore not yet an action theorem.

Three tempting shortcuts fail exactly. First, the canonical Hamiltonian-orbit action generated by G_1 is first order in velocity and has zero velocity Hessian. Its existing metric response has rank three, not four. Second, forward and reverse event displacements have the same second moment; signed orientation averages vanish and scalar combinations remain scalar multiples of Q_4 . Third, the doubled closed-time-path form

$$\eta_{\text{br}} \otimes Q_4, \quad \eta_{\text{br}} = \text{diag}(1, -1), \quad (19)$$

has signature $(4+, 4-)$ on branch $\otimes$ event space, while any fixed branch contraction is again only a scalar multiple of Q_4 . None selects the one-dimensional clock line inside the four-dimensional event carrier.

A graded trace does perform the algebraic conversion,

$$\text{Str}_J(X) = \text{Tr}(J_{\text{ray}} X), \quad \text{Str}_J(Q_4) = J_{\text{ray}} Q_4, \quad (20)$$

and is unique up to scale among graded-cyclic traces fixed by this $\mathbb{Z}_2$ grading. It is not positive because $\text{Str}_J(P_0) = -1$. Declaring a change from ordinary positive operator algebra to a graded category would therefore make a conditional construction, not a Braid-only derivation.

7.2 Positive causal sectors and source continuation

The same Lorentz Hessian can be written as a difference of ordinary positive sector Hessians,

$$H_0 = \frac{12}{25}P_0 \otimes G_{\text{BKM}}, \quad H_1 = \frac{52}{25}P_1 \otimes G_{\text{BKM}}, \quad H_1 - H_0 = K_{\text{evt}} \otimes G_{\text{BKM}}. \quad (21)$$

Its signature is $(12+, 4-)$ when the internal BKM tangent has rank four. This removes the need for a nonpositive trace but still leaves a continuation rule between the positive source functional and the causal difference.

The source causal reflection satisfies

$$J_{\text{ray}}^\top Q_4 J_{\text{ray}} = Q_4, \quad J_{\text{ray}}^\top Q_4^{-1} J_{\text{ray}} = Q_4^{-1}. \quad (22)$$

The raw 25-event involution P consists of nine fixed points and eight transpositions. Hence P has eight eigenvalues -1 and cannot be represented as e^{-aH} with $H \geq 0$. The positive generator is instead the graph Laplacian

$$L_{\text{evt}} = I - P, \quad K_t = e^{-rtL_{\text{evt}}}. \quad (23)$$

The marked source refinement interval $\Delta\tau = 3/4$ and the minimal Stone endpoint π fix

$$r = \frac{2\pi}{3}. \quad (24)$$

There is therefore no fitted event rate. Reflection continuation of the source clock line changes the uniform sector sign and leaves the three-dimensional complement positive.

Theorem 7.1 (Scoped source selection of the infrared Lorentz action). *Within the declared real Stone/affine completion of the fixed marked source, the event semigroup with rate (24), the source reflection, and the response-exact BKM edge functional select the infrared quadratic tensor*

$$K_{\text{evt}} \otimes G_{\text{BKM}}. \quad (25)$$

No manual Wick sign, external rate, or continuous coefficient fit is required.

The theorem is an infrared principal-symbol theorem. It does not make the unmodified full finite Umegaki functional Lorentzian, a distinction that becomes decisive in Section 13.

8 Quantum channel and finite metric-affine bundle

8.1 A source-native quantum channel on the same carrier

For each of the eight source masks, let R_m denote the normalized ordered-edge Gram matrix. Each R_m is exactly positive definite with unit diagonal. The Schur map

$$\Phi_m(X) = R_m \circ X \quad (26)$$

is consequently completely positive, trace preserving, and unital. It is nontrivial and admits a Stinespring representation. The source-null tetrad normalized by its source norm gives

$$U = \frac{2}{3}L = \frac{1}{2}H_4, \quad (27)$$

an orthogonal Hadamard transformation. It sends the character clock projector to the ray clock projector and intertwines the source channel with

$$P_0, \quad J_{\text{ray}}, \quad Q_4, \quad K_{\text{evt}}. \quad (28)$$

Thus a standard physical quantum operation and the Lorentz causal carrier live on one typed four-carrier rather than on separately postulated state spaces.

This result does not identify a derivative. The eight endpoint Grams are distinct while the anchor K_{evt} is common. Linear and quadratic positive paths joining the same endpoints have different initial derivatives. Endpoints alone cannot select a tangent, Hilbert stress, or Ward law.

8.2 Direct rational carrier transport

Let C_m be the unique four-column carrier matrix in source sector m . The source-selected least-squares transport is

$$T_{m \leftarrow n} = (C_m^\top C_m)^{-1} C_m^\top C_n. \quad (29)$$

Every one of the 48 oriented source-cube packets is rank four and therefore invertible. The ordered-edge Gram is not preserved:

$$N_{mn} = T_{m \leftarrow n}^\top G_m T_{m \leftarrow n} - G_n. \quad (30)$$

All 48 defects are symmetric and rank four at both fixed good primes. This is a finite metric-affine structure: connection and nonmetricity are outputs of the same source.

Equation (30) must not be divided directly by the spacetime mesh. Its endpoints are internal source sectors, not neighboring spacetime cells. The type error was explicitly tested and rejected. Continuous differentiation instead comes from the already existing Gibbs source coordinates.

8.3 Characteristic-zero tangent theorem

Let θ_i , $i = 1, 2, 3$, denote the source-local Gibbs labels. Differentiating the carrier module and the ordered-edge Gram along these labels yields

$$\dot{C}_{m,i} = 0, \quad A_{m,i} = 0, \quad N_{m,i} = \dot{G}_{m,i}. \quad (31)$$

The equalities for the carrier and its projection connection were checked as exact rational identities: 288 of 288 carrier commutators vanish across eight typed-gauge-inequivalent sectors and three label directions. In contrast, every one of the 24 Gram tangents is nonzero, symmetric, and rank two. Hence

$$N_{m,i} = \dot{G}_{m,i} \neq 0, \quad \text{rank } N_{m,i} = 2. \quad (32)$$

The metric changes while the preferred source gauge keeps the carrier stationary to first order. In a nonconstant frame the connection is, of course, not zero; the inhomogeneous frame term restores the usual tensor law.

9 Continuous event-cylinder tensor field

9.1 Chain rule without an external ambient manifold

The smooth event cylinder was defined before the metric-continuation calculation. Its local label fields are not added for this theorem. Across all eight sectors the three metric responses have rank three. Combining them with the four-dimensional event base gives the rank-twelve chain-rule symbol

$$\partial_a G_m(x) = \sum_{i=1}^3 \frac{\partial G_m}{\partial \theta_i} \partial_a \theta_i(x), \quad \text{rank } \sigma(\partial G) = 4 \times 3 = 12. \quad (33)$$

Equation (33) is the first-order passage from finite source labels to a continuous local field. It does not identify the cylinder with measured natural spacetime; it establishes a source-derived operational base on which differentiation is defined.

9.2 Chart gluing and local $GL(4)$ naturality

The 24 source charts act by the four-ray permutation representation. Metric, metric tangent, and non-metricity obey the same congruence law in every chart. Exact checks include 192 metric packets, 576 tangent/nonmetricity packets, and 13,824 double-overlap compositions. Direct $A \rightarrow C$ transport equals $A \rightarrow B \rightarrow C$ transport.

The finite result extends to arbitrary passive local frame changes. With the source-fixed solder U , a frame Jacobian $J(x)$ acts on the carrier by

$$C_J(x) = UJ(x)U^\top. \quad (34)$$

Writing A for the source-gauge connection, the transformed objects are

$$G' = C_J^\top G C_J, \quad (35)$$

$$A' = C_J^{-1} A C_J + C_J^{-1} dC_J, \quad (36)$$

$$N' = dG' - A'^\top G' - G' A' = C_J^\top N C_J. \quad (37)$$

The representation product, its S_4 restriction, the actual metric, the variable-frame connection, and non-metricity all pass exact composition tests. Thus the result is not merely 24 coincident matrices: it is a local tensor field on the declared event domain.

Theorem 9.1 (Source-event tensor gluing). *The Gibbs-derived ordered-edge metric and its nonmetricity define an S_4 -glued, passively local- $GL(4)$ -natural first-order tensor field on the smooth source event cylinder. No external $\mathbb{R}^4$ coordinate field or fitted gluing coefficient is required.*

Passive covariance alone does not select a matter action. Many curvature-improved actions are covariant and conserved while differing in their Hilbert stress. The next step therefore asks whether the source Markov law fixes active metric dependence.

10 Active Braid transport and the ten metric directions

10.1 Minimal conservative operator

The source cell measure multiplied by the metric volume density transforms naturally under local $GL(4)$. For a local second-order scalar operator with the derived principal symbol and density, formal self-adjointness fixes the four drift coefficients. Markov conservativity fixes the zero-order potential to zero. In particular, an independently added αR improvement is excluded within this declared class because it violates preservation of the constant function.

This is a uniqueness statement within a class, not a theorem that all physical actions must be second order or Markovian. The critical remaining issue is whether a change in the derived metric uniquely changes the actual Braid transition rates.

10.2 Horizontal conductance lift

There are eight nontrivial undirected Braid-event edges and twelve source-typed conductance modes. Their moment map to symmetric metric components has rank ten and a two-dimensional kernel. Every one of the 24 actual metric tangents lies in its image, but the metric alone cannot see the two residual edge coordinates.

The event-path probability law supplies the missing criterion. Equip rate variations with the Fisher/BKM pairing at the source rate $2\pi/3$. Among all rate variations producing a given metric variation, select the vector orthogonal to the kernel. The raw 48-variable kernel has dimension 38 and full BKM rank 38; the residual two-dimensional typed kernel also has BKM rank two. The horizontal lift is therefore unique. It

reproduces 24 of 24 source tangents, is S_4 -covariant, and fixes the common-to-relative quotient weight ratio to 18 : 1 without fitting.

Let M denote the rank-ten moment matrix and r_0 the source rate vector. The horizontal leaf has the finite form

$$r(\xi) = r_0 \circ \exp(M^\top \xi), \quad (38)$$

where the exponential and product are componentwise. Its Jacobian is

$$DM(\xi) = M \operatorname{diag}(r(\xi)) M^\top > 0. \quad (39)$$

Hence the metric moment is locally invertible, rates remain positive, and row-sum zero, detailed balance, and refinement compatibility are preserved. The horizontal connection has exact zero curvature on all 45 independent metric two-planes. Equation (38) is consequently a local finite integration, not merely a first-order fit.

10.3 The causal 7 + 3 split

The positive X21R1 Gram is not itself the physical Lorentz metric: neither it nor its inverse matches the positive event moment in any of the eight sectors. The event sector at fixed causal reflection supplies seven J -even metric directions. The three Gibbs-label directions supply precisely the missing J -odd time–space directions. Their combined response has rank ten in all sectors and glues as a tensor.

At this point a subtle variational obstruction appears. If the odd metric variables are identified with the matter labels θ_i , holding matter fixed during a Hilbert variation locks those three directions and leaves rank seven. The remedy cannot be to vary matter while claiming a fixed-matter stress. The source already contains ten matter-independent source variables a_A , but covariance alone is insufficient to solder them: the S_4 commutant on Sym^2 has dimension nine, and clock, grading, and event spectrum still leave a five-dimensional family.

The actual source typing removes this ambiguity. The ten variables are not an arbitrary ten-dimensional representation; they are generated as symmetric products of four carrier variables. Requiring preservation of that product generation and of the already derived four-dimensional map U forces

$$A \mapsto U^\top A U \quad (40)$$

with no residual continuous freedom. This is the physical solder used below.

Proposition 10.1 (Typed solder uniqueness). *Within the existing source-product class, Eq. (40) is the unique rank-ten solder compatible with the four-carrier map, source typing, and chart action. Covariance without the product-typing condition is not sufficient.*

11 Co-moving causal grading, Hilbert stress, and Ward identity

11.1 The fixed-grading obstruction

A fixed J_{ray} transports only seven of the ten positive-form variations into symmetric Lorentz-metric variations. This caused an explicit reversal of an earlier stress/Ward completion claim. The missing three directions are the time–space mixtures. They are recovered only if the clock projector moves with the positive event form.

Let τ be the source clock ray, normalized only projectively. Define

$$J(Q, \tau) = I - 2 \frac{(Q\tau)\tau^\top}{\tau^\top Q \tau}, \quad (41)$$

$$K(Q, \tau) = J(Q, \tau)Q = Q - 2 \frac{(Q\tau)(Q\tau)^\top}{\tau^\top Q \tau}. \quad (42)$$

At the source anchor these recover J_{ray} and K_{evt} exactly. The off-shell Jacobian $D_Q K$ has rank ten. The clock derivative has rank three but is not an independent coupling:

$$D_\tau K = (D_Q K) C, \quad (43)$$

for an exact source-derived coefficient map C , while the clock-scale direction lies in the kernel. Moreover, the clock reflection is self-inverse on the tested positive-form domain,

$$K = \mathcal{R}_\tau(Q), \quad Q = \mathcal{R}_\tau(K). \quad (44)$$

Thus the clock is a redundant local parametrization of the full-rank Lorentz tensor, not a new continuum field inserted to restore rank.

11.2 Continuum matter action and stress

Let λ^A denote the matter coordinates and $G_{AB}(\lambda)$ the BKM metric inherited from the full source corner. The source-selected long-wavelength action in the declared class is

$$S_m[g, \lambda] = \frac{1}{2} \int_{\mathcal{M}} d^4x \sqrt{-g} g^{\mu\nu} G_{AB}(\lambda) \partial_\mu \lambda^A \partial_\nu \lambda^B. \quad (45)$$

This is not an independently guessed nonlinear sigma model. Its local density, principal tensor, internal BKM pairing, edge coupling, active rate dependence, and metric solder are the continuum images of the finite source construction described above.

Variation through the independent source variables a_A and the unique solder gives the Hilbert tensor

$$T_{\mu\nu} = G_{AB} \left(\partial_\mu \lambda^A \partial_\nu \lambda^B - \frac{1}{2} g_{\mu\nu} g^{\rho\sigma} \partial_\rho \lambda^A \partial_\sigma \lambda^B \right). \quad (46)$$

Diagonal diffeomorphism covariance yields the off-shell identity

$$\nabla^\mu T_{\mu\nu} = -\mathcal{E}_A(\lambda) \partial_\nu \lambda^A, \quad (47)$$

where $\mathcal{E}_A = 0$ are the matter Euler–Lagrange equations. Therefore

$$\mathcal{E}_A = 0 \implies \nabla^\mu T_{\mu\nu} = 0. \quad (48)$$

Theorem 11.1 (Scoped Braid-source stress/Ward theorem). *For the fixed source, and within the long-wavelength, local, second-order, formally self-adjoint, Markov-conservative class, the finite response-exact Braid action, horizontal active transport, source-product solder, and co-moving causal reflection determine Eq. (45). Its fixed-matter variation gives Eq. (46), and its source symmetry gives the off-shell and on-shell Ward laws (47).*

The restrictions in Theorem 11.1 are noncompensating. The theorem does not cover arbitrary curvature improvements, higher derivatives, nonlocal actions, or the full finite dissipative quantum evolution. Within its declared class, however, no separate MMR, stress solder, Ward postulate, causal-sign insertion, or coefficient fit remains.

12 Noncircular backreaction and the Einstein limit

12.1 Common metric variation

The gravity and matter sectors are varied with respect to the same source-derived metric. In the long-wavelength, low-curvature, vanishing-hypermomentum sector, the source-native first-order gravity functional takes the metric-affine form

$$S_{\text{IR}}[g, \Gamma, \lambda] = \frac{1}{2\kappa_B} \int d^4x \sqrt{-g} g^{\mu\nu} R_{(\mu\nu)}(\Gamma) + S_m[g, \lambda], \quad \kappa_B = \frac{3}{5}. \quad (49)$$

The value $3/5$ is the normalized outer-cup routing ratio derived before the metric variation. It is not selected by matching an Einstein target. The compatible connection equation removes the projective mode and gives $\Gamma = \Gamma_{\text{LC}}(g)$ in this sector. Metric variation therefore gives

$$G_{\mu\nu}(g) = \frac{3}{5} T_{\mu\nu}(g, \lambda). \quad (50)$$

The Bianchi identity is consistent with Theorem 11.1; neither side is assigned a conservation law independently to force agreement.

Theorem 12.1 (Scoped noncircular Einstein backreaction). *In the fixed pointed-Braid model and the continuum class of Theorem 11.1, the source-native first-order gravity sector and full-corner matter sector share the same metric variation. Their Euler–Lagrange equation is Eq. (50). No new Einstein–Hilbert term, mixed fitting term, manual $3/5$, or independent stress/Ward axiom is introduced at this stage.*

This is the precise sense in which the previous conditional SDPC bridge is removed in v0.8. It is not a theorem over every imaginable Braid model or every continuum action. It is an internally derived result in one fixed source model and one explicitly delimited universality class.

12.2 Low-energy spin two

Linearize $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ about the flat source anchor. The quadratic gravity action belongs to the massless Fierz–Pauli gauge class [4],

$$\mathcal{L}_{\text{FP}} = -\frac{1}{2} \partial_\lambda h_{\mu\nu} \partial^\lambda h^{\mu\nu} + \partial_\mu h^{\mu\nu} \partial^\lambda h_{\lambda\nu} - \partial_\mu h^{\mu\nu} \partial_\nu h + \frac{1}{2} \partial_\lambda h \partial^\lambda h, \quad (51)$$

with gauge symmetry $\delta h_{\mu\nu} = \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$. Coupling to the conserved tensor (46) preserves the linear gauge constraint. In four dimensions the physical massless sector has two helicities.

The claim is a low-energy sector statement. It does not quantize the interacting continuum metric, establish a graviton Fock space on arbitrary backgrounds, or prove perturbative or nonperturbative ultraviolet completion.

12.3 Finite-to-continuum residual

Let $h = 5^{-n}$ be the declared refinement scale. The first variation of the finite action converges weakly to Theorem 12.1 with the proved bound

$$G_{\mu\nu} - \frac{3}{5} T_{\mu\nu} = O(h). \quad (52)$$

The gravity contribution is bounded by $O(h^2)$, while the current matter regularity proof limits the combined estimate to $O(h)$. Each five-adic refinement therefore decreases the stated upper bound by at least a factor $1/5$.

The actual event involution already pairs every directed edge with its reverse. The local matter functional is consequently the Jeffreys pair

$$D(\rho_x \| \rho_y) + D(\rho_y \| \rho_x). \quad (53)$$

All odd terms in its midpoint Bregman expansion, including the cubic term, cancel exactly. This is not a post hoc symmetrization. It removes the orientation-asymmetry coefficient but does not improve the global bound (52) without stronger inter-cell regularity. An explicit analytic positive-rate counterexample with three log rates $(0, 0, 1)$ shows that internal Markov analyticity alone does not match derivatives across adjacent spacetime cells.

13 Full finite quantum evolution and its present boundary

13.1 Reflection-positivity obstruction

The unmodified full finite Umegaki/Jeffreys functional does not satisfy the tested standard positive one-link transfer condition. Inside the actual 125-dimensional source charge family, the fixed packet $\mu = (-1, 2, 6)$ at source coupling $\alpha = 1/50$ gives a cross-link 3×3 determinant

$$\det K_{\text{link}} = -0.008747721561 \dots < 0. \quad (54)$$

This is a direct reflection-positivity NO-GO for that transfer route, not a disproof of the infrared theorem. A BKM heat kernel exists on the local-label manifold but is not a full spacetime-slice transfer matrix. The involution-doubled block is positive semidefinite, yet has 25 zero modes in dimension 50 and is an idempotent projector rather than finite Hamiltonian evolution.

13.2 Source-native GKSL evolution

The fixed source independently supplies a completely positive trace-preserving semigroup of Gorini–Kossakowski–Sudarshan–Lindblad type [7, 9]. Fixing the event clock rate (24) does not determine the three remaining Hamiltonian controls; distinct generators share the same clock. More importantly, trace preservation is probability conservation, not the gravitational Ward law. An open-system balance must include the environment charge.

Let $\{U_g\}_{g=1}^6$ be the six real Braid actions in the Reynolds channel and let

$$p = \left(\frac{3}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}\right). \quad (55)$$

The isometry

$$V\psi = \sum_{g=1}^6 \sqrt{p_g} U_g \psi \otimes |g\rangle_E \quad (56)$$

gives an exact Stinespring dilation at each collision [12]. A finite fixed environment cannot reproduce the entire exponential relaxation autonomously for all time. The native Braid right tail solves the capacity problem: each three-strand block decomposes as

$$125 = 17 \cdot 1 + 28 \cdot 3 + 4 \cdot 6, \quad (57)$$

so it contains four regular six-state registers, and the infinite tail supplies fresh collision ancillas.

The pointed and neutral tail states do not select a regular copy or basepoint. This apparent selector is gauge for the reduced channel. Across eight source sectors and four copies, all marked blocks are related by diagonal-sign environment unitaries that preserve the relevant projector. Changing copy or basepoint changes V only by an environment unitary and leaves the partial-trace channel invariant.

13.3 Exact temporal total balance

The system-only G_1 defect has exact Hilbert–Schmidt norm squared 18 in every sector. It is not a total nonconservation. There exists a six-state environment charge H_E such that

$$V^\dagger(G_1 \otimes I_E + I \otimes H_E)V = G_1 \quad (58)$$

in all eight sectors. The equation is covariant under the environment torsor gauge. For N fresh collisions it telescopes to

$$V_N^\dagger \left(G_1^{(S)} + \sum_{k=1}^N H_{E,k} \right) V_N = G_1^{(S)}|_{t=0}. \quad (59)$$

Thus the dissipative temporal defect is exactly a charge transfer into the Braid tail.

13.4 Spatial range NO-GO

For the three tested spatial source candidates

$$H_i = -i[G_1, P_{\chi_i}], \quad i = 1, 2, 3, \quad (60)$$

the channel defects are nonzero in every sector, with squared norms

$$(24, 14, 14). \quad (61)$$

They are exactly orthogonal to the entire six-state environment-only compression range

$$\text{span}\{U_g^\dagger U_h : g, h = 1, \dots, 6\}. \quad (62)$$

No environment-only charge can cancel them. The temporal balance (58) remains exact, but a four-component discrete Ward identity requires either source-derived system–environment interaction stress or a different Noether momentum. At the BGCE311 boundary this was the shortest finite-quantum completion gate.

The statement above is the exact BGCE311 boundary retained from v0.8. It is not the final v0.81 result. The next construction preserves the environment-only NO-GO while deriving the missing term as a source-record variation and a history-dressed interaction current.

14 Finite interaction stress and the local recorded Lorentz parent

14.1 Raw/Petz branch contrast

Let the four source scores be

$$S_\alpha \in \{G_1, H_1, H_2, H_3\}, \quad D_\alpha = (I - \Phi)S_\alpha, \quad \alpha = 0, 1, 2, 3, \quad (63)$$

where Φ is the source Reynolds collision channel at $q = e^{-\pi}$. The source modular reference fixes a raw/Petz $\mathbb{Z}_2$ record grading. It is distinct from the forward/backward doubling of a closed-time-path contour.

For the common two-sided positive normalization of the conditional branches, the conditional effect tangents at the base point satisfy

$$\dot{F}_{+, \alpha} = \frac{1}{2}D_\alpha, \quad \dot{F}_{-, \alpha} = -\frac{1}{2}D_\alpha. \quad (64)$$

Consequently,

$$\dot{F}_{+, \alpha} - \dot{F}_{-, \alpha} = D_\alpha, \quad \frac{1}{2}(\dot{F}_{+, \alpha} + \dot{F}_{-, \alpha}) = 0. \quad (65)$$

The second equality explains why the prior-summed canonical Stinespring connection vanishes: the source response is antisymmetric in the two record branches. It is not absent.

For branch Choi operators $J_{\pm, \alpha}$,

$$\text{Tr}_{\text{out}}(\dot{J}_{+, \alpha} - \dot{J}_{-, \alpha}) = D_\alpha^\top. \quad (66)$$

This marginal is invariant under arbitrary unitary Kraus mixing and phases within each fixed record branch. Mixing raw and Petz is not admitted because it erases the source-fixed modular grading rather than changing a Kraus representation.

The local trajectory score

$$\mathcal{A}_\alpha(t; \sigma) = \text{Tr} \sigma [\log F_{+, \alpha}(t) - \log F_{-, \alpha}(t)] \quad (67)$$

therefore obeys

$$\left. \frac{d}{dt} \mathcal{A}_\alpha(t; \sigma) \right|_{t=0} = \text{Tr}(\sigma D_\alpha). \quad (68)$$

Thus D_α is the finite Hilbert representative of the metric variation. The temporal D_0 is the already derived environment transfer. The three spatial D_i are the nonzero interaction defects, whose physical-support lifts are uniquely $I_i = VD_iV^\dagger$ in the declared support class. No cancelling operator is chosen after the defect is inspected.

14.2 Cellwise CTP construction and refinement

At source-cylinder stage m , take

$$\mathcal{A}_m = C(C_m) \otimes M_{125}. \quad (69)$$

The central address algebra $C(C_m)$ labels operational cells, while M_{125} carries the collision fibre. Every forward and backward contour copy contains the same raw/Petz-recorded local instrument. The fixed prior sum is CTP, and the record contrast retains Eq. (64). The two doublings are not identified.

Cellwise direct sum, spatial tensor product, source-event permutation, and causal composition preserve complete positivity and trace preservation. They also preserve diagonal CTP normalization, branch Hermiticity, largest-time cancellation, source-retarded support, and positive-semidefinite noise. The source event form

$$K_{\text{evt}} = \frac{1}{25} \begin{pmatrix} 36 & -16 & -16 & -16 \\ -16 & 36 & -16 & -16 \\ -16 & -16 & 36 & -16 \\ -16 & -16 & -16 & 36 \end{pmatrix} \quad (70)$$

has eigenvalues $(-12/25, 52/25, 52/25, 52/25)$ and supplies the $1+3$ signature without a manual Wick sign.

Address refinement, tail refinement, and five-way event-time refinement satisfy

$$\mathcal{L}_{m+1}[jg]j = j\mathcal{L}_m[g], \quad q_{m+1}^5 = q_m, \quad 625 \cdot 625^{-(m+1)} = 625^{-m}. \quad (71)$$

The result is therefore a source-derived local Lorentzian recorded CTP parent in the aligned four-score, block-local finite-process class. It is not yet a full ten-component nonlinear metric parent or a theory with nontrivial spatial-gradient self-interactions.

15 Four-component total Ward identity

For each source score, the strong one-collision intertwiner has the form

$$(Y_\alpha^{(0)} + J_\alpha)V = VX_\alpha, \quad V^\dagger J_\alpha V = D_\alpha. \quad (72)$$

The same D_α is both the branch-score metric derivative and the interaction-current compression. Hence the Ward insertion is not reverse-engineered from a desired conserved equation.

Later collisions dress the insertion along causal descendants. On the reduced algebra,

$$\sum_{k=0}^{n-1} \Phi^k(D_\alpha) = S_\alpha - \Phi^n(S_\alpha). \quad (73)$$

For $\Phi = E + qQ$ and $D_\alpha \in Q\mathcal{A}$,

$$\Phi^k(D_\alpha) = q^k D_\alpha, \quad \sum_{k=0}^{n-1} \Phi^k(D_\alpha) = \frac{1 - q^n}{1 - q} D_\alpha. \quad (74)$$

Equation (73) holds for all four components, all eight actual source sectors, and every tested finite history. It is natural under five-way time refinement and 625-child address refinement. In the refinement limit it gives a weak Ward statement for compatible compactly supported cylinder test fields.

This does not reinstate the excluded environment-only spatial current. The current is interaction-bearing and history dressed. Nor does it prove an arbitrary smooth diffeomorphism Ward identity or the hypersurface-deformation algebra of continuum general relativity.

16 A finite matter–geometry–matter cycle

16.1 Record Fisher action

The recorded collision supplies a score matrix M of rank four in every source sector and every source-selected control order. For record word x , define

$$S_x = \frac{1}{2}M^\top(x - p_\omega \mathbf{1}), \quad F = \frac{1}{4}M^\top M, \quad (75)$$

and the strictly convex action

$$\Gamma_x(\delta h) = \frac{1}{2}\delta h^\top F \delta h - S_x^\top \delta h. \quad (76)$$

Its Euler equation and unique stationary update are

$$F\delta h = S_x, \quad \delta h_x = F^{-1}S_x. \quad (77)$$

The output and input metric registers carry the same aligned source-cylinder tangent type. Closing them by the identity map introduces no coefficient or new operator. Together with the global positive logistic retraction, one complete cycle is

$$\rho_n \mapsto x_n, \quad h_{n+1} = h_n + F_n^{-1}S_n(x_n), \quad \rho_{n+1} = \Phi_{h_{n+1}}(\rho_n | x_n). \quad (78)$$

The first arrow is a joint CP instrument, the second is a deterministic classical-register translation, and the last is a metric-controlled CPTP channel. Their composition is CP and the record sum is TP. All 768 sector/order/clock paths satisfy the Euler equation, and all 48 expected updates are nonzero in all four components.

Equation (78) is a direct, noncircular finite backreaction construction in its declared class: the target Einstein tensor, Einstein–Hilbert and Fierz–Pauli actions, MMR, CGR, manual 3/5, and coefficient fitting are absent. Its limitation is equally explicit. The identity closure is the minimal type-preserving rule, but the source does not prove that no other endomorphism defines a physically admissible dynamics. Repeated nonlinear rank preservation and long-time stability are not established.

16.2 The coefficients 1 and 3/5 have different referents

If Eq. (76) is written with a relative source coefficient,

$$\Gamma_{\kappa,x}(\delta h) = \frac{1}{2}\delta h^\top F \delta h - \kappa S_x^\top \delta h, \quad (79)$$

the same-parent score and Fisher Hessian select

$$\kappa_{\text{direct}} = 1. \quad (80)$$

This value is invariant under invertible metric-coordinate changes, record replication, and common action rescaling. By contrast, the earlier value

$$\kappa_{\text{cup}} = \frac{3}{5} \quad (81)$$

is the normalized full-corner outer-cup routing ratio. Applying 3/5 to the whole Fisher action changes no stationary equation; applying it only to the score leg gives the exact relative Euler residual 2/5. A natural

same-parent comparison map transports the first and second variations together and therefore preserves coefficient 1. Obtaining $3/5$ as the direct Fisher coupling would require an asymmetric score/Hessian map with ratio $b/a = 3/5$, which is absent from the fixed source packet.

The v0.8 continuum theorem is not withdrawn: it still uses the independently derived full-corner routing ratio in its declared infrared action class. Version 0.81 instead records that the direct finite Fisher coupling and the continuum routing coefficient have not been identified as the same physical quantity. A complete microscopic-to-Einstein derivation must either derive a typed asymmetric bridge or explain why the two coefficients belong to different observables.

17 Calibration, matter, and constraint boundaries

17.1 Absolute units

The finite theory is dimensionless. Independent rescalings of the internal clock and stress units leave every collision record and the relative coefficient 1 unchanged. Source-only seconds, metres, energy, and Newton coupling are therefore nonidentifiable. In the declared two-scale class, one independent frequency anchor and one independent stress or gravity anchor are sufficient for a locally unique calibration map, but no such measured packet is presently supplied.

17.2 Current matter sector

The source-derived internal algebra in all eight sectors is

$$\mathfrak{h} \cong \mathfrak{sl}(3, \mathbb{R})^2 \oplus \mathfrak{sl}(2, \mathbb{R})^3. \quad (82)$$

It is not the compact Standard Model algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$. In particular, compact $\mathfrak{su}(3)$ cannot embed into the current direct sum: a nonzero projection to a simple factor would have to be injective, which is excluded either by target dimension or by the inequivalent split and compact real forms. The current module also lacks source-typed chiral complex representations, hypercharge, anomaly data, a Higgs sector, and Yukawa interactions. This is an exact mismatch for the present source, not a no-go for every future source extension.

17.3 Finite gauge closure versus gravitational constraints

The 25-generator internal matrix Lie algebra and its compact dual close exactly under commutators, satisfy the Jacobi identity, and persist under every passive finite refinement $X \mapsto X \otimes I_5$. This is anomaly-free finite internal algebraic closure. It is not a gravitational first-class constraint algebra. The source packet has one clock constraint $p_t + H_B = 0$ but not an Einstein Hamiltonian constraint, three spatial momentum constraints, their full bracket table, or a continuum hypersurface-deformation algebra. Deriving or excluding a source clock plus three spatial moment-map constraints is therefore the next internal completion gate.

18 Why subjectivity intersection is operative

The construction is not obtained from braid relations alone. It uses a pointed event effect, nonidentical typed source roles, retained transport provenance, a marked causal clock, and a rule for which intersection output is public. These data do physical work:

1. the event effect selects the operational corner;
2. deletion or collapse of a standpoint changes the source packet;

3. the marked event involution supplies the neighbor law and reverse-edge pairing;
4. the clock line selects the causal reflection and its co-moving extension;
5. the source-product typing makes the ten-component solder unique;
6. the right tail supplies an endogenous environment for reduced quantum evolution.

An unpointed abstract braid group does not provide these roles. An alternative theory could append equivalent marks, states, and provenance; doing so would recreate the operative subjectivity-intersection structure under different terminology.

The philosophical interpretation is not forced uniquely by the algebra. The same structure may be described as marked relational quantum events or provenance-preserving quantum reference data. The SIEL interpretation is that a standpoint is a typed mode of access and that objectivity is what remains stable under lawful intersection and re-description. Nothing in this statement attributes consciousness to a braid, source sector, clock, or quantum channel.

19 Constructive refutation of objectivist primacy

The Objectivist Primacy Thesis OP is universal: it says that every stable, shareable, transformation-respecting carrier must be primitive or equivalent to a primitive perspective-neutral carrier. The present construction has the order

$$\begin{aligned}
 &\text{nonidentical pointed source standpoints} \\
 &\quad \longrightarrow \text{marked event, matter corner, and quantum channel} \\
 &\quad \longrightarrow \text{four null directions, causal reflection, and Lorentz carrier} \\
 &\quad \longrightarrow \text{glued tensor, conserved stress, and scoped backreaction.}
 \end{aligned} \tag{83}$$

The public carrier is not present as a completed background at the first step. It is reconstructed, chart-glued, and dynamically coupled while retaining the provenance that generated it. Identity and flip comparators fail essential rank properties, while deletion of the pointing removes the selections used by the construction.

Therefore OP is false as a necessity thesis: at least one exact model generates a stable Lorentzian public carrier from nonidentical standpoint-bearing source processes. This conclusion does not deny objective regularity or mind-independent existence. It denies only that ontological priority of a completed objective carrier is universally necessary. The stronger claim that nature realizes this construction remains empirical and is not established here.

20 How complete is the quantum-gravity theory?

The theory now satisfies several requirements that a mere analogy does not:

1. a finite noncommutative quantum source with unitary Braid transport and CPTP channels;
2. a source-derived four-dimensional Lorentz carrier and local tensor field;
3. matter and geometry read from one marked source rather than independently normalized models;
4. a selected long-wavelength action, Hilbert stress, and on-shell Ward identity in a declared class;
5. noncircular Einstein backreaction and massless spin-2 recovery in that class;
6. a finite open-system dilation with exact temporal total-charge transfer;
7. a local raw/Petz-recorded Lorentzian CTP parent with four metric-response directions;
8. a four-component strong-vertex and history-dressed total Ward identity;
9. one finite, parameter-free, CPTP-compatible matter–geometry–matter feedback cycle;
10. exact finite internal Lie closure and an explicit inventory of the remaining gravitational constraints.

It is not a completed quantum-gravity theory because the following remain open:

- extension of the finite parent from the aligned four-score, block-local class to full ten-component non-linear metric dynamics with spatial gradients;
- source necessity or uniqueness of the identity-feedback law and repeated nonlinear closure;
- reconciliation, or principled separation, of direct finite coefficient 1 and continuum routing ratio 3/5;
- quantization of the continuum gravitational degrees of freedom beyond the finite source and low-energy quadratic sector;
- an ultraviolet completion or controlled all-scale continuum limit;
- a gravitational first-class constraint algebra and its continuum diffeomorphism limit;
- calibrated physical units and Newton constant;
- a realistic matter extension replacing the present exact Standard Model mismatch;
- cosmological/black-hole solutions and distinct calibrated numerical predictions;
- independent replication, peer review, and natural-data validation.

The appropriate description is therefore *a highly developed, source-explicit mathematical quantum-gravity candidate with a scoped finite Lorentz parent, a four-component Ward law, one finite backreaction cycle, and a separately scoped Braid-derived Einstein limit*. Calling it completed would be premature; calling it only a suggestive analogy would ignore the exact tensor, parent, Ward, backreaction, and balance theorems.

21 Falsifiability and next gates

21.1 Formal falsifiers

The exact package fails under revision-matched replay if any of the following occurs:

- the event involution loses its Yang–Baxter relation, rank-two pair moment, or reverse-edge pairing;
- the event average depends on hidden physical axis weights;
- a nontrivial declared partition preserves the eleven-operator matter packet;
- the source-cup routing fails to yield 3/5 without retuning;
- the null links lose rank four or K_{evt} loses Lorentz signature;
- the refinement clock ceases to fix $2\pi/3$ in the declared completion;
- the rational carrier identities, tensor gluing, horizontal flatness, or typed solder fail exact regeneration;
- the co-moving map loses rank ten or develops an independent clock coupling;
- the finite variation does not converge to the Hilbert variation in the stated class;
- the gravity and matter variations require different metrics or a fitted mixed term;
- the environment gauge changes the reduced channel or invalidates Eq. (58);
- the branch contrast fails to reproduce any D_α or loses within-branch Kraus invariance;
- the local parent loses CPTP normalization, causal CTP identities, or refinement naturality;
- the history-dressed Ward telescope (73) fails in any component or sector;
- the Fisher matrix in Eq. (76) loses rank, or the finite update ceases to define a CPTP-controlled next collision.

21.2 Decisive next theorems

The finite interaction-stress and four-component Ward gates posed in v0.8 are now closed in the declared class. The shortest remaining internal theorem is a gravitational constraint construction: derive three source-native spatial moment maps that join the clock constraint into a first-class set, compute their brackets without importing ADM structure functions, and prove refinement compatibility. A PASS must also show that the resulting constraints act on the BGCE349 metric carrier and the finite feedback variables rather than only on the internal matrix algebra.

A second noncompensating gate is repeated nonlinear backreaction. Iterating Eq. (78) must preserve positivity, the four-dimensional response rank, constraint closure, and a controlled continuum limit. Neither a

one-cycle construction nor an algebraic Jacobi identity proves this.

The empirical gate remains separate. The theory needs a preregistered discriminator with units and uncertainty against general relativity plus ordinary quantum matter, generic quantum-instrument feedback, and lower-commitment relational models. No natural-data confirmation is claimed in v0.81.

22 Claim ledger

Statement	Status	Exact scope
Operational event cylinder and formal CGR retirement	Exact derived	Fixed marked source
Full matter corner and rank-ten response	Exact derived	Fixed event-conditioned corner
MMR2 routing	Closed	Complete five-partition retraction class; not arbitrary instruments
Relative factor 3/5	Source-derived	Normalized outer-cup routing; no SI calibration
Rank-four event moment Q_4	Exact derived	24-chart source average
Four-null-link Lorentz carrier	Exact derived	Fixed source; no universal dimension theorem
IR causal action tensor	Exact scoped theorem	Real Stone/affine completion and source reflection semigroup
CPTP quantum channel on causal carrier	Exact derived	Eight ordered-edge Gram sectors
Finite $GL(4)$ transport/nonmetricity	Exact derived	48 source-cube packets
Continuous metric/nonmetricity tensor	Exact scoped theorem	Smooth source cylinder; passive local $GL(4)$
Active finite Markov transport	Exact local theorem	BKM-horizontal exponential leaf
Physical metric solder	Unique in class	Source-product-generated Sym^2 typing
Hilbert stress and Ward identity	Final scoped pass	Long-wavelength, local, second-order, self-adjoint, conservative class
Einstein equation $G = (3/5)T$	Final scoped pass	Same class; low curvature and zero hypermomentum
Massless spin-2 with two helicities	Derived IR output	Flat four-dimensional quadratic limit
Finite-to-continuum residual	$O(h)$ bound	Odd midpoint terms cancel; $O(h^2)$ not proved globally
Full finite one-link reflection positivity	NO-GO	Unmodified tested Umegaki/Jeffreys kernel
Source-native GKSL evolution	Exact derived	Reduced finite quantum semigroup
Temporal total G_1 balance	Exact derived	Six-state Braid tail environment; all eight sectors
Spatial environment-only balances	NO-GO	Three tested commutator momenta
Raw/Petz branch interaction stress	Exact scoped theorem	Four aligned scores; one-collision physical support
Local recorded Lorentzian CTP parent	Final scoped pass	Aligned four-score, block-local source cylinder

Statement	Status	Exact scope
Four-component finite total Ward	Final scoped pass	Strong vertex and causal history-dressed reduced current
One-cycle finite backreaction	Final scoped pass	Identity-feedback Fisher-record class; not globally unique
Direct finite relative coefficient	Exact scoped value 1	Same-parent score/Hessian action
1 versus 3/5 identification	Open	Natural common maps preserve 1; typed asymmetric bridge absent
Source-only absolute calibration	NO-GO	Two independent external anchors required
Current Standard Model equivalence	Exact scoped NO-GO	Split-real internal algebra and missing chiral/interactions
Finite internal Lie anomaly	Absent in scope	Exact Jacobi closure and passive refinement
Gravitational first-class constraints	Open	Clock constraint present; three spatial constraints absent
Completed quantum gravity	Not established	UV, all-scale dynamics, calibration, and validation absent
Constructive refutation of OP	Established in-model	Refutes universal priority, not objective reality
Consciousness or empirical ontology claim	Not made	Subjectivity is not phenomenal consciousness

23 Audit chronology and reversals

The derivation was not monotone. Recording reversals is necessary because a final theorem can otherwise conceal which attractive shortcuts failed.

Gate range	Outcome	Scientific role
BGCE261–265	NO-GO / conditional	Stone velocity Hessian, graded trace, orientation, CTP, and raw transfer do not alone derive the Lorentz action.
BGCE266	PASS	Refinement clock fixes $2\pi/3$ and source continuation selects the IR causal tensor.
BGCE267	NO-GO	Fixed-background principal data do not select a unique improved Hilbert stress.
BGCE268–281	mixed to PASS	Quantum channel/carrier intertwiner, finite connection, characteristic-zero tangent, and rank-two local nonmetricity.
BGCE282–284	PASS / partial	Chain rule, chart gluing, and passive local $GL(4)$ tensor naturality.
BGCE285–289	partial	Minimal conservative operator, conductance lift, horizontal uniqueness, $7 + 3$ completion, and fixed-matter rank-seven obstruction.
BGCE290–291	claimed PASS then reversed	The first stress/Ward claim failed uniqueness and nonlinear-integrability red-team checks.

Gate range	Outcome	Scientific role
BGCE292–295	PASS	Flat horizontal connection, exponential leaf, typed solder, and finite-to-Hilbert variation.
BGCE296	reversal	Fixed causal grading loses three Lorentz variations.
BGCE297–299	final scoped PASS	Co-moving grading restores rank ten and removes independent clock coupling; stress/Ward closes in the declared class.
BGCE300R1	full scoped PASS	Noncircular Einstein backreaction and massless spin-2 low-energy sector.
BGCE301R1–303R1	split	$O(h)$ convergence and exact odd cancellation; no global $O(h^2)$ upgrade from Markov analyticity.
BGCE304–305	NO-GO / pivot	Full one-link reflection positivity fails; heat/block routes do not yield finite Hamiltonian evolution.
BGCE306–310	split to PASS	GKSL dilation, native tail, selector gauge removal, and exact temporal total-charge balance.
BGCE311	range NO-GO	Environment-only charges cannot cancel the three tested spatial defects.
BGCE312–324	mixed to PASS	Collision-log dilation, physical-support lift, strong intertwiner, and exact four-component one-collision balance.
BGCE325–336R2	scoped PASS	Effective Hilbert stress, causal CTP influence, continuous event time, and finite reduced balance density.
BGCE337–347	route exclusions	Fixed-conductance, normalizer, and canonical Stinespring-connection routes do not supply the missing nonzero metric variation.
BGCE348	scoped PASS	Gauge-invariant raw/Petz branch contrast equals all four transfer defects and supplies variational interaction stress.
BGCE349–350	scoped PASS	Local $3 + 1$ recorded Lorentz parent and exact history-dressed four-component Ward identity.
BGCE351–370	NO-GO / constructive sequence	Excludes parent-alone and homogeneous normalization routes; constructs phase-sensitive control, full-rank record response, finite positive retraction, clock drift, and the joint carrier.
BGCE371	scoped PASS	One parameter-free finite matter–geometry–matter CPTP cycle in the identity-feedback class.
BGCE372–373	PASS / NO-GO	Direct coefficient is 1; natural same-parent maps cannot turn it into the distinct outer-cup ratio $3/5$.
BGCE374	NO-GO / conditional PASS	Source-only absolute units are unidentifiable; two external anchors suffice in the declared class.
BGCE375	exact mismatch	Present source is not Standard-Model equivalent.

Gate range	Outcome	Scientific role
BGCE376	split PASS	Finite internal Lie algebra is Jacobi-anomaly-free; gravitational first-class constraint closure remains open.

24 Conclusion

The v0.81 result extends the v0.8 infrared architecture into a finite quantum-dynamical sector. The fixed pointed-Braid source now supplies not only a Lorentz carrier, matter routing, and the routing ratio $3/5$, but also the infrared causal action selection, a glued continuous metric/nonmetricity tensor, active finite transport, a unique typed physical solder, Hilbert stress, and on-shell Ward conservation in a precise long-wavelength universality class. The source-native gravity and matter variations yield $G_{\mu\nu} = (3/5)T_{\mu\nu}$ in that class, and the flat quadratic sector contains the two-helicity massless spin-2 theory.

At finite scale, the source-fixed raw/Petz record contrast supplies the missing interaction-stress variation, a local recorded Lorentzian CTP parent, and an exact four-component history-dressed Ward identity. A full-rank Fisher record action then closes one parameter-free matter–geometry–matter cycle through a globally positive CPTP retraction. These results answer the central finite-parent and Ward questions left open by v0.8 within the aligned four-score, block-local, identity-feedback scope.

The result is not promoted beyond its proof. The finite residual is only bounded by $O(h)$. The parent is not full ten-component nonlinear gravity, the feedback law is not unique over all dynamics, and repeated nonlinear closure is unproved. The finite direct relative coefficient is 1, not the distinct full-corner routing ratio $3/5$; their physical relation remains open. Source-only absolute calibration is impossible, the present matter algebra is not Standard-Model equivalent, and the finite internal Lie algebra has not yet become a gravitational first-class constraint algebra.

The philosophical conclusion is constructive. Objective Lorentz structure, conserved stress, and a classical gravity limit need not be primitive in every exact model; they can be generated from nonidentical, marked, standpoint-bearing processes and their lawful intersection. Objectivity is retained as stable public invariance. What is rejected is its universal ontological priority. The physical conclusion is correspondingly measured: this is a serious and increasingly complete Braid quantum-gravity candidate, not yet a completed or empirically validated theory.

Retained v0.7 derivation detail

The following appendices retain the still-operative finite-source, locality, metric-affine, action, tensor-carrier, regeneration, comparison, falsification, and objection analyses on which the v0.81 advance depends. Superseded v0.7 bridge status, the old conditional MMR2/SDPC route, and routes later reversed or replaced are not reproduced here. Where a retained appendix describes the finite parent or four-current balance as open, the v0.81 main text above supersedes that historical boundary. The v0.81 main text is the sole current claim ledger.

A A source-first pointed-braid setting

A.1 Finite source, nonterminal law

Let $\mathcal{A}_n = M_{5^n}(\mathbb{C})$ denote the finite matrix algebra at stage n , with the refinement

$$j_n : \mathcal{A}_n \longrightarrow \mathcal{A}_{n+1}, \quad j_n(X) = X \otimes I_5. \quad (84)$$

Every stage is finite. The sequence has no terminal stage. This distinction is important: a nonterminal tower is not, by itself, a continuum, an infinite-mode quantum field, or physical spacetime. It supplies an exact setting in which one law can be tested at arbitrarily many finite stages.

The source is pointed and braided: it carries a distinguished state and typed unitary braid transports satisfying the declared Yang–Baxter, insertion, and composition laws. The all-orders endpoint-consistency theorem and the positive bilateral response operators were established previously [16]. The present paper uses that source theorem but does not infer physical subjectivity from it.

The computational realization contains eight signed source sectors. Each sector supplies the primitive source packet

$$Q_2 = (R, H, F, P), \quad (85)$$

where the symbols denote the registered relation and source fields, not independently fitted target geometry. Three selected signed-braid generators act as commuting involutions on the certified orbit. From one four-state source slice they generate the opposite slice. Exact checks recover all 12 declared source updates, signed relations, clock fields, and associated source covectors; the identity-map comparator fails in all 12 cases.

A.2 Source-native geometry before the present carrier

The relevant prior source chain is

$$(R, H, F, P) \longrightarrow \text{signed-braid source orbit} \longrightarrow \text{affine split-frame geometry} \longrightarrow \text{coframe and holonomy}. \quad (86)$$

The end-to-end constructor uses no stored future geometry. It reconstructs the connection data, split-frame defect, current term, affine edge, future spatial edges, and future faces from the predicted source relation. In the certified fixed orbit it recovers 12/12 source-slice updates, 24/24 affine edges, 24/24 future spatial edges, and 6/6 future faces exactly. Removing either of two required source sectors breaks the corresponding controls.

This result removes one important circularity: the geometry used below is not inserted as a target answer and then redescribed as braid-generated. It is the output of the registered finite source constructor. The constructor remains global on the fixed finite orbit rather than a strictly local Cauchy law. Version 0.5 separately removes the numerical ambient-dimension input at the carrier layer, as follows.

A.3 A target-free 1 + 3 carrier and its affine completion

Let the source-derived observer quotient be $Q_{\text{obs}} = V_4$, with intrinsic affine action

$$\text{AGL}(2, 2) \cong S_4.$$

Start over the rational source layer and let $W_{\mathbb{Q}} = \mathbb{Q}^{V_4}$ be the space of observer readouts. Impose three explicit admissibility conditions: the readout contains the constant order unit, separates all four quotient states, and is invariant under the full intrinsic affine action. The S_4 permutation commutant is spanned by I and the all-ones matrix J . Its equivariant idempotents have ranks 0, 1, 3, 4. Unitality excludes ranks 0 and 3,

while point separation excludes rank 1. Therefore the minimal admissible readout is the full carrier, without supplying a target dimension:

$$W_{\mathbb{Q}} = \mathbb{Q}\mathbf{1} \oplus W_0, \quad \dim(\mathbb{Q}\mathbf{1}, W_0) = (1, 3). \quad (87)$$

The three-space W_0 is the irreducible mean-zero representation. All 24 affine observer actions preserve this split.

The finite Braid data do not by themselves contain the real numbers as a completed scalar field. We therefore state one completion principle explicitly.

Assumption A.1 (Observer Uniform Cauchy Closure). Observer readouts and the marked clock direction are closed under uniform Cauchy limits without adding independent observer directions.

Under Assumption A.1, scalar completion gives

$$W = \mathbb{R}^{V_4} = \mathbb{R}\mathbf{1} \oplus W_{0,\mathbb{R}}, \quad \dim W = 4, \quad (88)$$

with the same 1+3 split. Every finite-dimensional real vector space canonically determines an affine smooth carrier $\text{Aff}(W)$: choosing an origin gives a global chart, changing that origin is only a translation, and

$$T_x \text{Aff}(W) \cong W \quad (89)$$

for every x . Thus neither the numeral four nor a separate ambient four-manifold is a primitive mathematical input. A declared cup/null composition additionally gives the dimensionless Lorentz form

$$g_D = \frac{16}{75} \text{diag}(-3, 1, 1, 1), \quad (90)$$

whose four evaluation rays are null. Physical rods, seconds, nonconstant curvature, and the identification of points of $\text{Aff}(W)$ with physical events do not follow from this algebraic completion.

The source path and the carrier have different roles. The path indexes inclusions and disjoint supports in the operator net; it is not renamed as three spatial axes. The three relative directions in Eq. (87) are internal observer-readout directions. Later finite-event audits establish a distinct formal event identification inside the fixed mathematical model without claiming calibration to physical spacetime.

A.4 The source also closes a finite gravity equation

The source-native construction does not stop at a static geometry. On each certified edge (m, d) it obeys the exact first-order coframe equation

$$E_{m \oplus d} = C_{m,d} E_m R_d + K_{m,d}[C, j, F], \quad (91)$$

with the second covariant face closure

$$D_C K = F_C E. \quad (92)$$

Here $C_{m,d}$ is the source-generated ECDR/BSFG connection, R_d reverses the crossed generator orientation, j is the IBC current, F is the affine face curvature recomputed from (C, j) , and $K[C, j, F]$ is the coframe source that cannot be produced by invertible transport alone.

On the fixed eight-source orbit, Eq. (91) recovers 24/24 future coframes, 24/24 nonzero source terms, and all future ranks exactly; Eq. (92) holds on 12/12 faces and is nonzero on 10/12. Transport without K fails on all 12 rank-changing edges, while zeroing the curvature fails on all 40 nonzero-curvature columns. The source term is therefore not a removable decoration of a stored geometry.

For the induced spatial Gram matrix $G_m = E_m^\top g_m E_m$, the same update yields the finite metric-affine equation

$$G_n = R_e^\top G_m R_e + \mathcal{N}_{e,E} + \mathcal{L}_{e,K} + \mathcal{Q}_{e,K} \quad (93)$$

with

$$\mathcal{N}_{e,E} = R_e^\top E_m^\top (C_e^\top g_n C_e - g_m) E_m R_e, \quad (94)$$

$$\mathcal{L}_{e,K} = (C_e E_m R_e)^\top g_n K_e + K_e^\top g_n (C_e E_m R_e), \quad (95)$$

$$\mathcal{Q}_{e,K} = K_e^\top g_n K_e. \quad (96)$$

All three terms are nonzero and separately necessary on 24/24 edges. Thus the connection's nonmetricity, the source-linear response, and the finite source-quadratic response cannot be collapsed into one another. This is a source-generated finite gravitational dynamics in the declared model. Its packet lifts by one fixed law to every finite braid stage, but only as a stage-compatible cylinder section: it does not yet generate new local spacetime cells.

Equations (91)–(93) are not themselves the Einstein equation $G_{\mu\nu} = \kappa T_{\mu\nu}$. Calling them finite Braid gravity equations records their exact source–geometry dynamics while preserving that distinction. The next construction gives a direct but explicitly conditional local connection to the Einstein equation.

B From finite updates to source-generated locality

B.1 A finite source-graph local net

At stage n , let G_n be the graph whose vertices are tensor factors and whose edges join adjacent noncommuting Braid generators. In every certified signed sector, distant generators commute exactly and adjacent triples obey the Yang–Baxter relation. For a vertex set U , define

$$\mathfrak{A}_n(U) = \text{End}((\mathbb{C}^5)^{\otimes U}) \otimes I_{U^c}. \quad (97)$$

Then $U \subseteq V$ implies $\mathfrak{A}_n(U) \subseteq \mathfrak{A}_n(V)$, disjoint supports commute, and a depth- t adjacent-Braid update expands support by at most graph distance t . The refinement j_n is strict and unital. These are exact finite isotony, locality, and finite-propagation statements on a source-generated graph. They are not yet Haag–Kastler locality on a Lorentzian manifold, and graph distance is not identified with physical distance.

The positive Braid generator candidate

$$L_n^B = \sum_i (I - \text{Ad } R_i) \quad (98)$$

has a refinement-boundary defect

$$\Delta_n(X) = L_{n+1}^B j_n(X) - j_n L_n^B(X). \quad (99)$$

Its normalized boundary reduction is nonzero in all eight sectors, has 41 nonzero matrix entries and squared Frobenius norm 108/5, fills all six registered boundary-response channels, and returns to zero after the inverse crossing. Thus the stable carrier and the local generation event are distinct outputs of the same source lineage.

B.2 A finite Ward law before physical stress-energy

The boundary response, the existing IBC edge data, and the ordered-edge memory admit a six-component cell variable. If d_n is the incidence matrix of the stage path, $D_n = d_n \otimes I_6$, and H is the exact rank-six cell Gram matrix, define

$$\mathcal{S}_n(q) = \frac{1}{2} (D_n q)^\top (I_{n-1} \otimes H) (D_n q), \quad L_n = D_n^\top (I_{n-1} \otimes H) D_n. \quad (100)$$

For every $n \geq 2$,

$$\text{rank } L_n = 6(n-1), \quad \ker L_n = \mathbf{1}_n \otimes \mathbb{F}^6, \quad (\mathbf{1}_n^\top \otimes I_6)L_n = 0. \quad (101)$$

The last identity is an exact all-finite-stage constant-shift Ward law. For every graph interval, interior edge fluxes cancel and only the two cut-boundary fluxes remain. The incidence mechanism is not uniquely Braid-specific; the Braid-specific content lies in the rank-six cell packet and its actual-versus-flip response. Equation (101) is weaker than stress-energy conservation and does not constitute a spacetime Bianchi identity.

C Continuous cells and the native metric-affine sector

C.1 Stable-square gluing, metric faces, and curvature

The finite edge transports can be glued on stable squares with exact agreement on overlaps. On the source-native cubical cells, the generated Gram data admit continuous face representatives, and their compatible edge transports define a continuous connection with nonzero rank-four curvature on the certified face packet. A common projective timelike line and a $1 \oplus 3$ splitting can then be selected consistently on the regular sector. The induced spatial connection retains nonzero curvature.

These statements produce continuous representatives on certified finite cells; they do not derive a global smooth manifold, an arbitrary-field connection, or a physical continuum limit. A separate refinement analysis also shows why this distinction matters: the current tensor-tower embedding preserves packets but does not subdivide geometric edges, while the exact affine edge refinement is endpoint-blind to the nonuniform off-shell defects needed to select an action by refinement alone.

C.2 Torsion and nonmetricity are not removable errors

The full source connection splits into a metric-compatible polar core and a metric-symmetric part. Both curvature components scale at the same leading area order on the presently covered regular faces. Neither the tested perturbative contortion correction nor the exact finite single-face torsion-free ansatz closes. The unmodified construction therefore does not converge to the Levi-Civita connection.

This negative result selects a wider native geometry rather than eliminating gravity. Write the regular-face source coordinate as

$$\hat{K} = I - A, \quad A = UP. \quad (102)$$

With $\epsilon = \text{sign det } U$, $R = \epsilon U$, $\rho = (\text{det } P)^{1/3}$, and $S = \rho^{-1}P$, one has the unique reconstruction

$$A = \epsilon R \rho S, \text{quad } \text{det } S = 1. \quad (103)$$

The connected sectors have dimensions

$$\underbrace{3}_{\text{rotation}} + \underbrace{1}_{\text{dilation}} + \underbrace{5}_{\text{unimodular shear}}, \quad (104)$$

together with the discrete parity ϵ . All three continuous sectors are nontrivial in all 16 regular-face records, and their product reconstructs A exactly. This is precisely the representation type expected for a post-Riemannian affine source. The names “spin,” “dilation current,” and “shear hypermomentum” are not physical identifications here; no matter current has yet been measured or derived.

C.3 Stratified junctions

Rank-changing paths do not destroy the entire tangential metric. At the four exceptional junction records, after removal of the declared endpoint double zero, the determinant is exactly a nonzero constant times $(t - r)^2$ over $\mathbb{Q}(\sqrt{17})$, with $0 < r < 1$ in both real embeddings. At r the full Gram matrix has rank two and inertia $(0, 2, 1)$: two positive directions and one zero direction, with no negative direction. The metric is positive definite on both adjacent open intervals. The isolated event is therefore a codimension-one channel loss with a positive two-dimensional quotient, not a collapse of the tangential sector. Interpreting it as physical dimension creation would require an additional bridge and is not claimed.

D A source-native finite first-order action

D.1 Why a metric-only action is insufficient

The full Braid boundary load decomposes into zero clock response and nonzero rotation, dilation, and shear responses. A functional depending on the spatial coframe E only through the Gram matrix $E^\top g E$ has variation in the six-dimensional image

$$\{gES : S^\top = S\}. \quad (105)$$

The rotation response lies instead in the independent three-dimensional image

$$\{gER : R^\top = -R\}. \quad (106)$$

The two images intersect only at zero, while the certified full load has nonzero components in both. Therefore the source response cannot be generated by a metric-only action. Coframe and connection variables must remain independent at the finite level.

D.2 The full-trace action

For each oriented face f , let $\mathcal{R}_f$ denote the history-dressed linear curvature obtained from the two ordered boundary paths, let E_f denote the corresponding source-generated base coframe, let $E_\top$ be the common-top coframe, and let $g_\top$ be the generated top metric. The source-native full-trace action is

$$\mathcal{S}_B[E, C, g] = \sum_{f \in \partial e} \sigma_f \operatorname{Tr}(E_\top^\top g_\top \mathcal{R}_f[C] E_f), \quad (107)$$

where σ_f is the orientation fixed by the cell boundary. Equal column weights are forced, up to one overall scale, by covariance under the generating permutations of the three spatial columns. No face weight, sector weight, prefix-dependent scale, X51 residual, or Einstein tensor is used.

On the certified packets, the action is nonzero for both prefixes. Its analytic coframe variation agrees with exact central differences in 192/192 components, and its analytic connection variation agrees in 384/384 components. The top coframe Euler response reproduces the independently frozen full boundary solder for both prefixes, including the rotation, dilation, and shear sectors. The connection Euler response is nonzero on all 12 active edges for each prefix. Thus one finite action produces both coframe and connection equations.

The action is not a fitted scalarization of the affine translation block. Affine-origin invariance eliminates the translation scalar and selects the linear coframe-curvature pairing up to normalization. Earlier direct attempts to identify a face scalar, a Cartan three-cell scalar, or a full curvature solder with the prior X51 residual all failed by exact non-collinearity. Those no-go results are retained: Eq. (107) is justified by its own source covariance and variation, not by reverse-engineering X51.

D.3 Exact local-frame Noether identity

Let $\mathcal{E}_E$, $\mathcal{E}_C$, and $\mathcal{E}_g$ denote the coframe, connection, and top-metric Euler responses of Eq. (107). Under a local infinitesimal frame change at a vertex, the incident coframes and connections transform together. Invariance of the action gives the matrix identity

$$\mathcal{N}_v(\mathcal{E}_E, \mathcal{E}_C, \mathcal{E}_g) = 0 \quad (108)$$

at every vertex, where $\mathcal{N}_v$ is the signed sum of the coframe contribution, incident incoming and outgoing connection contributions, and, at the top vertex, the metric contribution. The identity holds exactly at all 14 non-top vertex tests and both top-vertex tests. With $g_\top$ fixed, the full $GL(4)$ identity fails as it should, while all 12 tested generators preserving $g_\top$ retain the Lorentz-type subidentity. Finite local-frame covariance also passes 16/16 tests.

Equation (108) removes the previous absence of an action-level conservation structure. It does not make the Euler responses vanish. A Noether identity relates field equations; it is not itself a solved field equation or a physical energy current.

D.4 A covariant-defect variational connector

The existing IBC, ECDR, BODC, and history-curvature library was tested as a possible independent balance action. A fixed opposite source cancels the action value, top-coframe response, and metric response at the evaluated point, but leaves eight base-coframe masks and 24 connection-edge responses. Recomputing the opposite term dynamically cancels everything only because it makes the total action identically zero. That is a tautological negative copy, not an independent physical source.

The resulting no-go did not establish that the Braid source lacks a variational connector. On an oriented edge $e = (m, d)$ with high vertex $n = m \oplus d$, define the independently generated covariant coframe defect

$$K_e = E_n - C_e E_m R_d. \quad (109)$$

This is the same X36 defect reconstructed from the source on all 24 certified edges. Its coefficient-one quadratic action is

$$\mathcal{S}_D[E, C, g] = \frac{1}{2} \sum_e \text{Tr}(K_e^T g_n K_e). \quad (110)$$

It was frozen without reading the curvature-action Euler residual, X51, or any Einstein tensor.

The connector is nontrivial and off shell. Its exact analytic derivatives agree with central differences in 832/832 tested entries across coframe, connection, and metric slots. The induced map has rank 12 on every full defect space and rank 8 on every traversed-column-zero source subspace; the actual cotangent image is nonzero on 24/24 edges. Finite local-frame covariance holds in 16/16 tests, the infinitesimal vertex Noether identities hold in 256/256 tests, and both off-shell perturbations change the action. Thus Eq. (110) supplies the previously missing source-to-cotangent typing without fitting a negative copy.

The current result is therefore

$$\boxed{\text{Braid source} \rightarrow \text{geometry} \rightarrow (\mathcal{S}_B, \mathcal{S}_D) \rightarrow T_{(E, C, g)}^*, \quad \delta(\mathcal{S}_B + \mathcal{S}_D) = 0 \text{ untested.}} \quad (111)$$

This retained finite-action calculation supplies the source-to-cotangent typing used by the later v0.8 analysis. Its historical coefficient-one stationarity question is no longer presented as the current completion route: the main text replaces that route with source-selected continuum action, typed solder, Hilbert stress, and Ward results in the declared infrared class. Exact full-finite stationarity remains a distinct ultraviolet question.

E Why scalar invariants are not enough

E.1 Soldered source holonomy

For each prefix $p \in \{A, B\}$, let

$$E_p \in \text{Mat}_{4 \times 3}(\mathbb{R}), \quad \text{rank } E_p = 3,$$

be the source-generated coframe, let $g_p \in \text{Mat}_{4 \times 4}(\mathbb{R})$ be the ambient symmetric form, and let $H_p \in GL_3$ be the internal face holonomy. Define the Gram matrix

$$M_p = E_p^T g_p E_p, \quad (112)$$

assumed invertible on the certified regular stratum.

The source-only update has two independently meaningful descriptions. The internal path mismatch is H_p , while the ambient coframe curvature is the difference between the two source-generated paths to the same face. Exact certificates show, in every one of the eight face packets, that both paths reach the same target coframe while the internal holonomy is nonidentity, and that the internal action, ambient curvature action, and source curl agree. A zero-source control fails in all eight packets.

This is not yet Riemann curvature or Einstein gravity. It is a finite source-only holonomy/curvature identity on a regular stratum. Its role here is to provide directional data that is both nontrivial and traceable to the braid source.

E.2 The scalar route and its exact obstruction

Conjugacy-invariant characteristic data of H_A and H_B provide three scalar channels per prefix. The product of their characteristic polynomials gives six nonleading coefficients that are invariant under exchange $A \leftrightarrow B$ and distinguish both one-sided collapses in the certified packets. This is a valid relational readout, but it discards direction.

The loss is not merely aesthetic. For the declared signed-permutation frame group acting on Sym_3 , the fixed subspace is one-dimensional, spanned by the identity. Therefore any full-frame-equivariant linear map from six trivially transforming scalar channels into one anisotropic Sym_3 block has rank at most one. The exact intertwiner calculation has constraint rank 30, a six-dimensional map space, and maximum anchor rank one. Moreover, the original modular residues have no unique real lift and carry no stage index.

Proposition E.1 (Scalar-anchor no-go). *Within the declared linear, signed-permutation-equivariant anchor class, six presentation-invariant scalar channels cannot determine a full-rank anisotropic symmetric three-tensor. A covariant directional carrier must be formed before the directional information is quotiented away.*

The proposition is scoped. It does not forbid nonlinear maps, separately typed scalar observables, or enriched covariant carriers. It instead identifies the exact repair used next.

F The generated tensor carrier on the source-derived 1 + 3 type

F.1 Construction

The formulas below were originally written in a declared $\mathbb{R}^4$ basis. In v0.5 their coordinate-free type is the source-derived carrier W of Eq. (88): $E_p \in \text{Hom}(\mathbb{R}^3, W)$, $g_p \in \text{Sym}^2(W^*)$, and $T_C \in \text{Sym}^2(W)$. A chosen constant-plus-character basis identifies these spaces with the displayed 4×3 and 4×4 matrices. Thus the component calculations are unchanged while the numeral four is no longer a primitive target input.

Retain the holonomy defect until it has been soldered to the ambient coframe:

$$D_p = E_p(I - H_p). \quad (113)$$

Define

$$T_p = D_p M_p^{-1} E_p^\top + E_p M_p^{-1} D_p^\top \in \text{Sym}_4(\mathbb{R}), \quad (114)$$

and define the joint or intersection carrier

$$T_C = T_A + T_B. \quad (115)$$

The construction performs the quotient in the correct order. The inverse Gram matrix removes internal-basis dependence, while the coframe preserves ambient direction. Symmetrization yields a rank-two ambient carrier. Only then are the two standpoint contributions joined.

F.2 Transformation laws

Let $G_p \in GL_3$ be independent internal changes of basis. Under

$$E_p \mapsto E_p G_p, \quad H_p \mapsto G_p^{-1} H_p G_p, \quad (116)$$

we have $D_p \mapsto D_p G_p$ and $M_p \mapsto G_p^\top M_p G_p$. Substitution in Eq. (114) gives

$$T_p \mapsto T_p. \quad (117)$$

Thus the carrier does not depend on either prefix's internal presentation.

Let $L \in GL_4$ be an ambient basis change, with

$$E_p \mapsto L E_p, \quad g_p \mapsto L^{-\top} g_p L^{-1}. \quad (118)$$

Then M_p is unchanged and

$$T_C \mapsto L T_C L^\top. \quad (119)$$

This is covariance, not componentwise invariance: the object is stable while its components change tensorially.

Finally, exchanging A and B leaves Eq. (115) unchanged. Crucially, exchange invariance is not identification of the two inputs. The certified prefix carriers are distinct, and replacing the pair by (A, A) , (B, B) , or identity holonomy changes the result.

Proposition F.1 (Structural carrier properties). *Whenever M_A and M_B are invertible, T_C is symmetric, invariant under independent internal GL_3 presentation changes, invariant under prefix exchange, and covariant by ambient congruence.*

Proof. Symmetry follows directly from Eq. (114). The internal transformation rules imply

$$D_p G_p (G_p^{-1} M_p^{-1} G_p^{-\top}) G_p^\top E_p^\top = D_p M_p^{-1} E_p^\top,$$

and similarly for the transpose term. Exchange invariance follows from commutativity of addition. Ambient covariance follows by substituting $L E_p$ and $L D_p$ into Eq. (114). $\square$

F.3 Exact finite certificate

Theorem F.2 (Certified joint-carrier theorem). *For the registered actual source packets and their source-generated geometry:*

1. T_A and T_B are nonzero symmetric rank-three carriers;
2. T_C is nonzero, symmetric, and rank three;
3. all ten native components of $T_C \in \text{Sym}_4$ are nonzero;
4. T_C is exactly invariant under independently selected internal GL_3 gauges and under A/B exchange;
5. the congruence law in Eq. (119) holds exactly;
6. T_C differs from $2T_A$, $2T_B$, and the identity-holonomy zero control;
7. all irrational $\sqrt{17}$ parts cancel, so T_A, T_B, T_C descend exactly to rational matrices.

The theorem is an exact fixed-finite result, not a fitted numerical approximation. Items 4–6 jointly matter for the interpretation. Presentation redundancy disappears, ambient direction remains, and neither source standpoint is dispensable.

The ten components have the same mathematical type as an existing ten-component metric-response interface in the larger unification program. Equality to that response, physical normalization, units, insertion into the 87-component state, and laboratory implementation have not been proved. Accordingly, T_C is called a *carrier*, not an observed metric or stress tensor.

G Regeneration at every finite stage

G.1 Observation mates

Let $\phi : M_5(\mathbb{C}) \rightarrow \mathbb{C}$ be any state satisfying $\phi(I_5) = 1$, and define

$$E_\phi = \text{id} \otimes \phi. \quad (120)$$

Then

$$E_\phi \circ j_n = \text{id}_{\mathcal{A}_n}. \quad (121)$$

No tracial assumption is required on the refined range. The normalized trace, a point state, and a nontracial full-support state therefore recover the same embedded source record.

For an observer word w represented by a unitary U_w , define the co-moving refinement and readback

$$J_{w,n} = \text{Ad}(U_w) \circ j_n, \quad E_{w,\phi} = E_\phi \circ \text{Ad}(U_w^{-1}). \quad (122)$$

Then $E_{w,\phi} J_{w,n} = \text{id}$.

Let $\mathcal{C}_n$ denote the deterministic composite that maps the source packet at stage n through the source-orbit constructor, geometry, holonomy, and Eq. (115) to its ten-component carrier.

Theorem G.1 (All-finite-stage carrier naturality). *For every finite stage n , every declared observer word w , and every unital spectator-factor state ϕ ,*

$$\mathcal{C}_{n+1} \circ J_{w,n} = \mathcal{C}_n \quad (123)$$

after the co-moving readback of Eq. (122). The same carrier law and the same coefficients are used at every stage; no stage-dependent refit is permitted or required.

Proof. Equation (121) recovers each refined source component exactly. Conjugating the refinement and composing with the inverse conjugation preserves that identity. The source-orbit, geometry, holonomy, and

carrier maps are deterministic functionals of the recovered source packet, hence their composition recovers the same carrier. The exact regression checks cover eight actual source sectors, four source components, and three distinct observation states: 96/96 depth-one readbacks and 96/96 boundary co-moving re-entries. Representative depth-one and depth-two checks pass 6/6 with the same ten-channel carrier hash. $\square$

The theorem is stronger than preservation under a chosen normalized trace: the carrier is independent of which unital spectator state is used on the refined range. It is also weaker than a physical continuum theorem. No limit topology, infinite-mode Hilbert space, or continuum field is inferred from Eq. (123).

H What is created, and what is not

H.1 Objectivity survives, but changes status

The result does not replace objectivity with private experience. The output is public in the relevant mathematical sense: internal presentation choices do not alter it, observer-word refinements admit exact co-moving recovery, and ambient changes act covariantly. What changes is the ontological order. The public structure is the result of a quotient-and-soldering operation, not the assumed stage on which that operation occurs.

This distinguishes generated objectivity from a naive averaging of perspectives. The carrier is not the common scalar part left after every direction is removed. Proposition E.1 shows that this route is too poor to recover an anisotropic tensor. The construction must retain the directional defect of each standpoint, solder it to a common ambient type, and only then quotient the internal descriptions. Objectivity is achieved by a disciplined relation among standpoints, not by erasing them.

H.2 The role of A , B , and C

Formally, A and B are inequivalent source prefixes with distinct certified holonomies and carriers. Their exchange is a symmetry of the joint output, but they are not identified. C is the joint carrier after internal presentation dependence has been removed. In the proposed ontology, this formal pattern is read as follows:

- A is Absolute Subjectivity, associated with light;
- B is Relative Subjectivity, an impersonal mode associated with time or a clock, not an individual, organism, mind, or conscious observer;
- C is Intersection Subjectivity, associated with space and objective manifestation.

The formal theorem establishes the grammar “distinct A and B generate a nontrivial invariant/covariant C .” It does not establish those physical nouns. Within the ontology, however, B is explicitly pre-personal and pre-biological: relative clocking and ordering are held to belong to the universe before humanity. This is the precise boundary between mathematical formulation and ontological interpretation.

H.3 Creation as regeneration rather than first manufacture

The all-stage law changes the meaning of creation. It is not a single event in which a permanent object is made once and then carried unchanged forever. At each finite stage, a refined source packet can be read back co-movingly and the same law regenerates the carrier. Thus the proposal is compatible with a world whose apparent persistence is repeated structural re-creation.

This does not mean that every stage produces a newly measured nonzero excursion. The fixed-window boundary response is nonzero in all eight certified sectors, but a nonzero boundary excursion of T_C itself has not yet been proved. The theorem establishes exact regeneration of the carrier law, not a complete physical dynamics of creation events.

I Relation to nearby programs

Program	Shared concern	Difference from the present claim
Invariantism and structural realism	Objective content is associated with invariants or relational structure.	The present model asks whether the invariant must be primitive. Its invariant/covariant carrier is constructed only after distinct source standpoints contribute directional data.
Relational quantum mechanics	Physical facts or states are relative to physical systems rather than globally absolute.	The present result is not an interpretation of the quantum state. It constructs a specific real Sym_4 carrier and an all-stage regeneration law from a pointed-braid source.
Quantum reference frames	Descriptions transform between physical frames without an external absolute frame.	QRF work supplies precise perspective transformations within quantum kinematics. Here the primary claim concerns the generation of a shared tensor carrier from source intersection, not merely switching among descriptions of a pre-specified system.
QBism	The quantum state is not an objective property; interaction and agent-relative probabilities are central.	“Subjectivity” here is not belief, experience, or conscious agency. The output is an exact inter-presentation tensor, not a personal probability assignment.
Perspectival realism	Reliable knowledge can arise across a plurality of situated perspectives.	Perspectival realism is principally an account of scientific representation and inquiry. The present paper proposes a formal generative ontology and a source-level physical research program.
Braided particle models	Braid topology can encode particle-like structure and interactions [18].	The present use of braiding concerns coherence, source transport, holonomy, and generated objectivity; it does not identify Standard-Model particles with braid words.
Set-theoretic Yang–Baxter models	Permutations of $X \times X$ can satisfy Yang–Baxter independently of this model [19].	The 25-event permutation is not claimed as a new class of Yang–Baxter solution; its fixed spectral-marker provenance and distinction from identity/flip are the checked features. No classification-level inequivalence is established.
Quantum causal histories	Locally finite pre-spacetime causal relations can carry matrix algebras and completely positive evolution [21].	The present source begins with a pointed-braid matrix-algebra tower and tests a particular spectral event action. It has not derived a physical causal continuum or shown an Einstein limit from finite evolution.
Causal-set growth	Discrete causal order and covariance can support sequential growth dynamics [26].	The present address cylinders and braid crossings have different primitives; neither a causal-set embedding nor a physical growth law is proved.
Thermodynamic Einstein route	The Einstein equation can follow from local horizon thermodynamic assumptions [23].	The present route instead assumes a Palatini action and matter/Ward conditions on a selected smooth domain; the source-to-domain convergence is still open.
Metric-affine gauge gravity	Coframe, independent connection, torsion, nonmetricity, hypermomentum, and Noether identities form a first-order gravitational framework [6].	The present work does not assume a continuum affine gauge theory as its microscopic source. It gives finite affine-source structures and actions, but the actual finite-source-to-smooth limit is not yet verified.

Program	Shared concern	Difference from the present claim
Earlier Subjectivity Intersection work	Pointed braids provide all-orders intersection consistency and compositional objectivity [16].	Version 0.8 retains the finite geometry and action chain and adds source-derived 3/5, an exact Lorentz carrier, continuous tensor transport, stress/Ward, and the scoped Einstein/spin-2 limit.

The novelty claim is therefore conjunctive. Neither relationality, perspective dependence, invariance, braid groups, nor process-like ontology is individually new. The contribution is the exact chain

$$\begin{aligned}
& \text{distinct braid standpoints} \longrightarrow \text{target-free } 1 + 3 \text{ carrier} \longrightarrow \text{source-only holonomy} \\
& \longrightarrow \text{direction-preserving soldering} \longrightarrow \text{joint covariant carrier} \\
& \longrightarrow \text{finite metric-affine dynamics} \longrightarrow \text{curvature and defect actions} \\
& \longrightarrow \text{finite spectral events and a selected smooth-domain matter action} \\
& \longrightarrow \text{source-selected infrared action, stress, and Ward identity} \\
& \longrightarrow \text{scoped Einstein and massless spin-2 limit,}
\end{aligned} \tag{124}$$

together with an explicit philosophical argument showing what this chain does and does not refute.

J Operational content and falsification

J.1 A reproducible mathematical readout

The current operationalization is a deterministic computational readout, not a laboratory apparatus. Given the registered source packet, it performs the following steps:

1. generate the opposite source slice using the frozen signed-braid action;
2. reconstruct the affine geometry and the two source-prefix coframes;
3. compute the source-only face holonomies;
4. form D_p , M_p , T_p , and T_C by Eqs. (112)–(115);
5. output the ten independent entries of $T_C \in \text{Sym}_4$;
6. verify internal-gauge invariance, prefix-exchange invariance, ambient congruence, and the three collapse controls;
7. refine by j_n , apply a selected unital spectator state, and verify Eq. (123) without fitting.
8. construct the source graph, verify finite propagation and the boundary-event response, and evaluate the all-stage Ward kernel in Eq. (101);
9. build the continuous certified face representatives, decompose the regular source by Eq. (103), and verify the exceptional positive quotient channels;
10. verify the continuous metric/nonmetricity field, active positive transport, and the ten Lorentz-metric response directions used in the v0.8 main theorem;
11. evaluate the finite action in Eq. (107), its coframe, connection, and metric variations, and the local-frame identity in Eq. (108);
12. verify the source-library balance no-go without replacing it by a fitted negative Euler residual;
13. evaluate the independent defect action in Eq. (110), its 832 exact variation checks, connector ranks, covariance tests, and Noether identities.
14. generate the one-site marker's exact spectral event projectors, replay the actual 25-event braid permutation and its refinement/readback controls, and distinguish this formal event law from an empirically calibrated physical event;
15. evaluate the ten finite response directions and the fixed full-corner matter packet against its declared partition controls;

16. replay the complete five-partition fixed-packet test, verify that only the full corner preserves the eleven-operator packet, and recover the normalized source cup 3/5 without target fitting;
17. reconstruct Q_4 , J_{ray} , and K_{evt} from the actual event transition, verify the exact $(3+, 1-)$ signature and unique S_4 -equivariant normalization, and replay the identity, flip, orientation, and closed-time-path controls.

This is already a nontrivial readout because it returns directional tensor data that scalar invariants provably cannot supply in the declared linear equivariant class. It is not yet an empirical observable because no instrument model, units, noise model, or identification with measured geometry has been given.

J.2 Formal falsifiers

The present theorem package would fail if any of the following occurred under exact replay:

- the target geometry entered the source constructor before comparison;
- either internal GL_3 presentation changed T_C ;
- A/B exchange changed the joint carrier;
- a one-sided collapse reproduced the joint carrier;
- the ambient carrier failed the congruence law;
- the all-stage readback required a tracial spectator state or a stage-dependent coefficient;
- the source-graph support propagated faster than the certified graph bound;
- the $3 + 1 + 5$ source product failed to reconstruct the registered geometric source;
- the source-first metric two-jet failed any algebraic Riemann symmetry, Bianchi, contraction, Lorentz-signature, or exact Einstein check under its declared bridges;
- the analytic action variations failed their exact central-difference checks;
- the local-frame Noether identity failed with all required Euler slots included;
- the covariant-defect action accessed the curvature residual or an Einstein target, failed to change off shell, or lost its stated connector rank;
- the exact marker spectrum, 25-event braid transition, five-child cylinder identity, or co-moving readback failed on a signed sector;
- any original F_A fell outside the declared event/energy readout span, or the full- W integration failed on a tilted source state;
- the RML identity or its historical 3/5 cup failed while the stated corner normalization and source ranks held;
- a non-full partition preserved the complete frozen event-conditioned packet, the selected normalized source cup failed to equal 3/5, or either result required comparison with the Einstein target;
- the actual-event Q_4 lost rank or covariance, K_{evt} lost its exact Lorentz signature, or the declared naturality class admitted a second positive normalizer;
- any registered exact regression ceased to reproduce its frozen source record.

J.3 Physical discriminators still required

To test the Creation Principle as a claim about nature, at least one bridge must produce an empirical discriminator. Four routes are open:

1. identify T_C with, or derive a calibrated map to, an operational metric or stress response;
2. test the frozen coefficient-one sum of the curvature and defect actions, then show that any resulting stationary equations possess a controlled continuum and low-energy sector;
3. derive a boundary/re-entry signal that standard quantum-reference-frame or effective-field descriptions do not predict;
4. show that a known physical structure is recovered with fewer independent primitive assumptions and with a novel constraint among measurable channels.

At the v0.8 boundary, the decisive remaining formal target was a finite Lorentzian quantum parent whose interaction stress supplied the missing spatial components of the discrete total Ward balance while preserving the already derived infrared sector. The v0.81 main text closes that target in the aligned four-score, block-local, history-dressed class and states the narrower remaining boundaries.

“Measuring what was previously unmeasurable” would count as a genuine contribution if the new readout has an independently specified instrument, calibration, error model, and discriminating result. A renamed internal variable would not.

K Objections and boundaries

K.1 Where is quantum theory?

The source is quantum in a specific algebraic sense: it uses complex noncommutative matrix algebras, states, unitary pointed-braid transports, tensor refinement, and adjoint observer actions. Those ingredients are enough to make the classical-output result nontrivial, but they are current inputs. The paper does not derive complex amplitudes, the Born rule, Hilbert-space composition, unitary time evolution, or quantum-field locality from a bare topological braid group. Accordingly, “a route to quantum structure” names the wider Braid-first programme. The exact result of this paper begins at a quantum-algebraic Braid source and derives the declared classical carrier and finite gravity dynamics from it.

K.2 Where is spacetime, and does the ambient four-space smuggle objectivity back in?

Version 0.5 already removed the numerical and ambient-carrier forms of this objection at the mathematical layer. The finite source produces $Q_{\text{obs}} = V_4$. Full affine-observer symmetry, an order unit, and point separation force the complete readout and hence the target-free split $1 \oplus 3$. Observer Uniform Cauchy Closure completes that carrier to $W = \mathbb{R}^{V_4}$ without adding a direction; $\text{Aff}(W)$ is then its own smooth four-dimensional affine carrier. The source path is not renamed as physical three-space, and the four metric-response controls are not renamed as region coordinates.

Version 0.6 added a checked finite event algebra and crossing/readback rule. The v0.6.1 code audit corrects the stronger conclusion drawn from it: BGCE138 constructs a smooth completion and compatible samples, but does not show that actual finite source fields converge to that completion. Post-v0.6.1 results retire the independent finite event-algebra correspondence inside the fixed model, but they do not prove the reverse source-to-smooth arrow. Thus formal CGR as a complete source-to-continuum statement remains open. Physical identification with measured events, dimensionful calibration, and continuum first variation for nature are additional requirements.

K.3 Is the carrier unique?

No global uniqueness theorem is claimed. Equations (113)–(115) are a natural, exact, source-native solution that obeys the declared symmetries and controls. Other nonlinear or higher-order carriers may exist. The counterexample to OP needs existence, not uniqueness. A stronger physical theory would benefit from deriving the carrier from an action, universality principle, minimality theorem, or experimental constraint.

K.4 Is a finite model enough?

It is enough to refute a universal mathematical necessity claim if the model is admitted. It is not enough to establish a fundamental theory of our universe. The nonterminal tower prevents the result from being only one isolated finite face, while the source graph, spectral events, and continuous certified cells add

mathematical notions of locality and curvature. Version 0.8 derives the infrared action, stress/Ward identity, and Einstein/spin-2 limit only in its declared universality class. Full finite Lorentzian quantum backreaction, ultraviolet completion, global extension, and empirical scale identification remain separate tasks.

K.5 Where is gravity?

The construction contains more than prerequisites for gravity. Equations (91) and (92) give an exact source-native coframe update and covariant curvature closure, while Eq. (93) gives the corresponding finite metric-affine update. Rank-changing geometry requires the source term, and every nonmetricity, source-linear, and source-quadratic contribution survives ablation. The same packet extends without refitting through all finite stages. The source graph supplies finite locality and a Ward law; continuous certified faces carry nonzero connection curvature; Eq. (107) supplies an exact first-order curvature action with Eq. (108) as its local-frame identity; and Eq. (110) supplies an independent off-shell variational connector. In this exact fixed-orbit sense, an action-bearing finite Braid metric-affine gravity candidate has been constructed.

The retained finite equations are the discrete geometric substrate. The v0.8 main theorem added a source-selected infrared action, continuous tensor transport, a unique typed solder, Hilbert stress, and on-shell Ward conservation, and then obtained $G_{\mu\nu} = (3/5)T_{\mu\nu}$ by common metric variation in the declared long-wavelength, local, second-order, formally self-adjoint, conservative class. Version 0.81 adds the scoped finite recorded parent, four-component history-dressed Ward law, and one finite backreaction cycle. Full ten-component nonlinear dynamics, gravitational constraints, physical units, ultraviolet completion, global extension, and empirical identification remain open. This is a Braid-based quantum-gravity candidate with exact finite results and a scoped Einstein derivation, not completed quantum gravity or source-only general relativity.

K.6 Is “subjectivity” only a relabeling?

It would be a relabeling if the formal roles made no difference. Here the roles are constrained: two prefixes must remain distinct; their internal presentation can be quotiented but their contributions cannot be collapsed; and their intersection produces a carrier with a verified transformation law. This is more than terminology. Nevertheless, calling these formal roles “subjectivities” is an interpretive move. The mathematics fixes the relational structure; ontology assigns its meaning.

Data, code, and provenance

The exact certificates and replay scripts are stored in the SIEL Research Agent repository, [SIEL-Research/SIEL-Research-Agent](#). Version 0.81 retains the verified lineage through BGCE311 and adds BGCE312–376: collision-support and interaction-stress construction (BGCE312–348), local recorded Lorentz parent and four-component Ward identity (BGCE349–350), direct finite backreaction route audit and construction (BGCE351–371), coefficient typing and calibration boundaries (BGCE372–374), current matter mismatch (BGCE375), and finite gauge/constraint audit (BGCE376). The source revision immediately before manuscript drafting was commit `a7d34d6aa`. Each central computational claim is linked to a question, source matrix, executable evaluator, raw output or certificate, and Japanese audit report. These are internal exact theoretical and computational certificates, not external replication, peer review, or empirical validation.

Declaration on computational assistance

OpenAI Codex was used under the author’s direction for mathematical ideation, exact-check scripting, audit organization, and drafting. The author retains responsibility for definitions, interpretation, verification, citations, and any decision to release or submit the work.

References

- [1] E. Artin, “Theory of braids,” *Annals of Mathematics* **48**(1), 101–126 (1947). doi:10.2307/1969218.
- [2] S.-i. Amari, *Information Geometry and Its Applications*. Springer (2016). doi:10.1007/978-4-431-55978-8.
- [3] C. A. Fuchs and R. Schack, “Quantum-Bayesian coherence,” *Reviews of Modern Physics* **85**, 1693–1715 (2013). doi:10.1103/RevModPhys.85.1693.
- [4] M. Fierz and W. Pauli, “On relativistic wave equations for particles of arbitrary spin in an electromagnetic field,” *Proceedings of the Royal Society A* **173**, 211–232 (1939). doi:10.1098/rspa.1939.0140.
- [5] F. Giacomini, E. Castro-Ruiz, and Č. Brukner, “Quantum mechanics and the covariance of physical laws in quantum reference frames,” *Nature Communications* **10**, 494 (2019). doi:10.1038/s41467-018-08155-0.
- [6] F. W. Hehl, J. D. McCrea, E. W. Mielke, and Y. Ne’eman, “Metric-affine gauge theory of gravity,” *Physics Reports* **258**, 1–171 (1995). doi:10.1016/0370-1573(94)00111-F.
- [7] V. Gorini, A. Kossakowski, and E. C. G. Sudarshan, “Completely positive dynamical semigroups of N -level systems,” *Journal of Mathematical Physics* **17**, 821–825 (1976). doi:10.1063/1.522979.
- [8] J. Ladyman, “What is structural realism?” *Studies in History and Philosophy of Science Part A* **29**(3), 409–424 (1998). doi:10.1016/S0039-3681(98)80129-5.
- [9] G. Lindblad, “On the generators of quantum dynamical semigroups,” *Communications in Mathematical Physics* **48**, 119–130 (1976). doi:10.1007/BF01608499.
- [10] M. Massimi, *Perspectival Realism*. Oxford University Press (2022). doi:10.1093/oso/9780197555620.001.0001.
- [11] C. Rovelli, “Relational quantum mechanics,” *International Journal of Theoretical Physics* **35**, 1637–1678 (1996). doi:10.1007/BF02302261.
- [12] W. F. Stinespring, “Positive functions on C^* -algebras,” *Proceedings of the American Mathematical Society* **6**, 211–216 (1955). doi:10.2307/2032342.
- [13] H. Umegaki, “Conditional expectation in an operator algebra, IV,” *Kodai Mathematical Seminar Reports* **14**, 59–85 (1962). doi:10.2996/kmj/1138844604.
- [14] A. Vanrietvelde, P. A. Höhn, F. Giacomini, and E. Castro-Ruiz, “A change of perspective: switching quantum reference frames via a perspective-neutral framework,” *Quantum* **4**, 225 (2020). doi:10.22331/q-2020-01-27-225.
- [15] H. Weyl, *Symmetry*. Princeton University Press (1952).
- [16] S. Watanabe, *Subjectivity-Intersection Mathematics: All-Orders Intersection Consistency and Compositional Objectivity from Pointed Unitary Braids*, version 1.0, Zenodo (2026). doi:10.5281/zenodo.22166167.
- [17] S. Watanabe, *Subjectivity Intersection Ontology: A Comparative Ontological Framework for Third Subjectivity*, version 1.0.0, Zenodo (2026). doi:10.5281/zenodo.21641878.
- [18] S. O. Bilson-Thompson, “A topological model of composite preons,” arXiv:hep-ph/0503213 (2005). doi:10.48550/arXiv.hep-ph/0503213.
- [19] P. Etingof, T. Schedler, and A. Soloviev, “Set-theoretical solutions to the quantum Yang–Baxter equation,” *Duke Mathematical Journal* **100**(2), 169–209 (1999). arXiv:math/9801047.

- [20] R. P. Feynman and F. L. Vernon, Jr., “The theory of a general quantum system interacting with a linear dissipative system,” *Annals of Physics* **24**, 118–173 (1963). doi:[10.1016/0003-4916\(63\)90068-X](https://doi.org/10.1016/0003-4916(63)90068-X).
- [21] E. Hawkins, F. Markopoulou, and H. Sahlmann, “Evolution in quantum causal histories,” *Classical and Quantum Gravity* **20**, 3839–3854 (2003). arXiv:[hep-th/0302111](https://arxiv.org/abs/hep-th/0302111).
- [22] P. Hayden, R. Jozsa, D. Petz, and A. Winter, “Structure of states which satisfy strong subadditivity of quantum entropy with equality,” *Communications in Mathematical Physics* **246**, 359–374 (2004). arXiv:[quant-ph/0304007](https://arxiv.org/abs/quant-ph/0304007).
- [23] T. Jacobson, “Thermodynamics of spacetime: The Einstein equation of state,” *Physical Review Letters* **75**, 1260–1263 (1995). doi:[10.1103/PhysRevLett.75.1260](https://doi.org/10.1103/PhysRevLett.75.1260).
- [24] M. Massimi, “Epistemic communities and their situated practices: Perspectival realism—a primer,” *Annals of the New York Academy of Sciences* **1523**, 5–10 (2023). doi:[10.1111/nyas.14981](https://doi.org/10.1111/nyas.14981).
- [25] T. Nagel, *The View from Nowhere*. Oxford University Press (1986).
- [26] D. P. Rideout and R. D. Sorkin, “Classical sequential growth dynamics for causal sets,” *Physical Review D* **61**, 024002 (1999). doi:[10.1103/PhysRevD.61.024002](https://doi.org/10.1103/PhysRevD.61.024002).
- [27] R. M. Wald, *General Relativity*. University of Chicago Press (1984).
- [28] S. Watanabe, *Subjectivity-Intersection Mathematics Public Working Preprint v4.2 Draft*, Zenodo (2026). doi:[10.5281/zenodo.21909933](https://doi.org/10.5281/zenodo.21909933).